Product Market Power as a Labor Distortion: the Case of Home Production Substitutes

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This version: July 10, 2026

Abstract

We introduce monopolistic competition in markets for home production substitutes into a model of household time allocation and study the implications for female labor supply. Households in our model consume market goods and home-produced goods; the latter are produced using a combination of the wife's time and home inputs, where the market for home inputs is monopolistically competitive. The resulting markups on home inputs are endogenous and depend on both the elasticity of substitution between products and the home production technology. Markups directly raise the price of outsourcing housework and thereby reduce female labor supply; we show theoretically that markups can also amplify or dampen the effect of other economic forces, such as the cost of home inputs and the gender wage gap. We estimate our model using a combination of time allocation, expenditure, and markup data. Our estimates imply that endogenous markups amplify the labor supply effects of technological progress in housework substitutes by approximately 12%. The framework identifies a distinct channel through which product market power shapes female labor supply, operating through household time allocation.

Keywords: Home production, market power, female labor supply, gender inequality

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Introduction

Since at least Reid (1934) and Becker (1965), economists have understood that household production uses a combination of households' own time with market-purchased inputs. Buying a cleaning product or hiring a housekeeping service substitutes a market input for the household's own time, and the optimal extent of this substitution depends on the price of the input relative to the opportunity cost of time. Because women have historically borne, and continue to bear, the majority of home production,[1] this "make or buy" decision is central to understanding female labor supply.

A large literature has studied the forces that shift this margin and evaluated their importance in explaining observed aggregate trends, specifically the large gender convergence in labor force participation (Goldin 2014; Olivetti and Petrongolo 2016) and the contemporaneous decrease in the time spent in home production (see e.g. Aguiar and Hurst (2007), Bridgman (2023), Bridgman and Valentinyi (2026)). Technological progress that lowers the cost of home production substitutes has been identified as a primary driver of this convergence (Greenwood, Seshadri, and Yorukoglu 2005; Albanesi and Olivetti 2016). The narrowing gender wage gap also raises the opportunity cost of home time, further drawing women into market work (Jones, Manuelli, and McGrattan 2015). Both forces operate by making market substitutes relatively cheaper, tilting the household's make-or-buy decision toward buying.

A common feature of this work is that the markets for home production substitutes are almost always treated as perfectly competitive: in particular, cost reductions – such as those due to technological progress – pass through to prices in full. In this paper, we show theoretically and quantitatively that the market structure of these industries matters. Imperfect competition in markets for home production substitutes leads to markups on these products above marginal cost, which constrain the household's ability to marketize home production and thereby directly distort labor supply. Moreover, we argue that imperfect competition, and the endogenous product markups it generates, may interact in nontrivial ways with the other aforementioned forces that impact labor supply – such as cost-reducing technological progress or the falling gender wage gap – either dampening or amplifying them.

The theoretical contribution is to introduce monopolistic competition in home production substitutes into a model of household time allocation, following a similar approach to Albrecht, Phelan, and Pretnar (2025). The starting point is a simple and standard model of household production (see, e.g., Greenwood, Guner, and Vandenbroucke (2017)). The

[1] See e.g. Aguiar and Hurst (2007) and Hancock, Lafortune, and Low (2025) for evidence on the gender gap in home production time.

household consists of a husband, who works in the market full time, and a wife, who divides time between market work and home production; there is an exogenous gender gap in wages when working in the market. The household consumes market goods and home-produced goods where the latter can be produced using a combination of home production time and household inputs. The crucial departure from this traditional framework is that we assume imperfect competition in both market goods and home production substitutes. On the market good side, we assume that the household has preferences over a continuum of varieties, each produced by a monopolistic firm as in Dixit and Stiglitz (1977). On the home production side, we assume that the household consumes a continuum of varieties of home goods, where each variety can in turn be produced using a combination of time and an input supplied by a monopolistic firm. The model is analytically tractable and nests the workhorse framework in family economics as well as standard applications of the Dixit and Stiglitz (1977) monopolistic competition framework. The markup on home production substitutes is endogenous and depends on the elasticity of substitution between time and inputs within a variety, as well as the elasticity of substitution across home input varieties. This endogenous markup creates a wedge that distorts equilibrium time allocation. Both market good markups and home substitute markups distort the allocation of time between home production and market work, but through qualitatively different channels: the former acts as a sales tax on market consumption (the usual channel examined extensively in the literature, see e.g. Edmond, Midrigan, and Xu 2023), while the latter raises the cost of outsourcing home production.

The main theoretical result is that the endogenous response of the markup can either amplify or dampen the labor supply effects of the exogenous forces that the literature has emphasized: the narrowing gender wage gap and the falling marginal cost of home production substitutes. The direction depends on whether different varieties of home substitutes are more substitutable with each other than any variety is with the household's own time. If the elasticity of substitution across varieties is greater than the elasticity of substitution between time and home inputs, then the endogenous markup response amplifies the effects of exogenous changes. Technological progress that lowers the cost of home inputs makes households more reliant on purchased substitutes; since different brands are highly substitutable, firms face more elastic demand and lower their markups, so the price falls by more than the exogenous cost reduction. Similarly, a narrowing gender wage gap raises the opportunity cost of home time, pushing households toward purchased substitutes and triggering the same markup decline. In both cases, the endogenous markup response reinforces the direct effect on female labor supply. The opposite occurs if the elasticity of substitution across varieties is smaller than the elasticity of substitution between time and home inputs: in this case, the pass-through from costs to prices is less than one-for-one, and the endogenous markup acts as a dampening force.

The quantitative contribution is to estimate the model and use it to assess the implications of market power in the home inputs sector. Estimating the model's parameters requires data on household time allocation, household expenditures, and product markups; and moreover, requires expenditures and product markups to be distinguished by product category (market goods vs. home inputs). We accomplish this through a new measurement strategy that combines several datasets: the NielsenIQ Consumer Panel, the American Time Use Survey (ATUS), the Consumer Expenditure Survey (CEX), and the Current Population Survey (CPS). A key step in our measurement strategy is the classification of products into either market goods or home substitutes in a manner consistent across datasets. In our baseline analysis, we treat housework (interior cleaning and laundry) as our measure of home production, though our approach is applicable more generally. We show that all the parameters of the model are identified using the available data. Our estimation of the household production technology extends the approaches of Fang and Yang (2022) and Fang, Hannusch, and Silos (2020), with the crucial difference being the presence of product markups, and therefore the necessity of merging household expenditure and time allocation data with the NielsenIQ price data.

Our estimates imply that different brands of housework substitutes are far more substitutable with each other than any one brand is with the household's own time. Despite this high substitutability across input varieties, the model is consistent with the observation that markups on home inputs are of similar magnitude to those on market goods: the elasticity of substitution across varieties is not a sufficient statistic for the markup on home inputs, which depends also on the substitutability between inputs and household time.

This estimated ranking of the elasticities of substitution implies, as explained above, that endogenous markups *amplify* the effects of both cost reductions and wage convergence on female labor supply. A fall in the physical cost of housework substitutes is amplified by approximately 12% because the endogenous markup decline reinforces the direct cost reduction. This suggests a reinterpretation of the observed fall in the prices of home substitutes documented, e.g., in Greenwood, Seshadri, and Yorukoglu (2005). In our estimated model, any given reduction in prices requires a less than one-for-one reduction in the marginal cost; conversely, a policy intervention, e.g. a subsidy on these products, would have more than a one-for-one effect on their equilibrium price. Our results also have policy implications. First, they suggest that regulation of market power in home substitute markets can serve as an additional, and largely overlooked, policy lever in promoting female labor force participation, as emphasized in Sanders (2024). Second, they also imply that the effects of either direct subsidies on home inputs or policies narrowing the gender wage gap can be amplified through the response of home input markups.

Related Literature. This paper studies the connection between product market power and household time allocation and shows how to operationalize this connection both theoretically and quantitatively. We are motivated by a vast literature in family macroeconomics, which has identified time-saving technology and markets as major contributors to increasing female labor force participation. This research documents how falling costs of home production substitutes, from home appliances and electrification to childcare and infant formula, have reduced the time burden of home production and enabled greater female labor market participation (see e.g., Greenwood, Seshadri, and Yorukoglu, 2005, Attanasio, Low, and Sánchez-Marcos, 2008, Cavalcanti and Tavares, 2008, Coen-Pirani, León, and Lugauer, 2010, Dinkelman, 2011, and Albanesi and Olivetti, 2016). A related strand of this literature also shows that such technological progress helps explain broader demographic trends including marriage, divorce, and fertility patterns (Greenwood and Guner, 2008, Greenwood, Guner, Kocharkov, and Santos, 2016, Santos and Weiss, 2016, and Bar, Hazan, Leukhina, Weiss, and Zoabi, 2018, and Cerina, Moro, and Rendall, 2021). All of these studies treat the relevant product markets as perfectly competitive, implying, in particular, that the falling costs of these products are passed through to prices one-for-one. Relative to this literature, our contribution is to study the effect of market power for home inputs.[2] We argue that deviating from the perfectly competitive benchmark in these markets has important implications. First, the trend in prices of home inputs should no longer be interpreted as stemming from marginal costs only. Instead, the trend in markups of these products has to be taken into account. Second, market power interacts with other driving forces such as the falling gender wage gap: the prices of home inputs respond to demand through monopolistic pricing, so exercises that change wages while holding prices fixed are misspecified. Third, the analysis opens the door to additional policy avenues, such as regulation of markets for home inputs, for impacting female labor supply.

On the theoretical side, our framework is closest to the innovative work by Albrecht, Phelan, and Pretnar (2025), who likewise allow household time to be substitutable with market-purchased goods and show that this has important implications for the inefficiency of markup heterogeneity. We see our work as highly complementary, addressing a different

[2] While some work has certainly documented market power in specific home input markets, such as childcare (Bastos and Cristia, 2012 and Berlinski, Ferreyra, Flabbi, and Martin, 2024), infant formula (Wang, 2024), and processed foods (Nevo, 2001), we undertake a macro approach by considering market power for a group of such products, and our focus is on its effects on female labor supply rather than consumer welfare in each specific market.

research question, and providing a quantitative implementation. In Albrecht, Phelan, and Pretnar (2025), the focus is on how households' ability to use time as a substitute affects markups on products and the resulting allocation of resources across firms. Conversely, the focus of our analysis is on how market power for certain products affects the household's allocation of time. Unlike Albrecht, Phelan, and Pretnar (2025), we explicitly interpret these products as home production substitutes and focus on the effect on female labor supply. Most importantly, we propose and implement a measurement and calibration strategy to estimate the model and quantify this product market power channel.

On the methodological side, our quantitative approach builds on and extends the existing literature on estimating the household production technology. Our estimation approach is similar to Fang and Yang (2022) and Fang, Hannusch, and Silos (2020) in using cross-sectional variation in expenditure shares by household income to identify key parameters: namely, the elasticity of substitution between time and home inputs in the household production function, and the elasticity of substitution between market and household-produced goods. The crucial difference is the inclusion of product market power, which adds a third key parameter – the elasticity of substitution between varieties of home inputs – and necessitates the use of markup data to identify it. We provide a constructive identification proof, which shows that the identification of the home technology from time use and expenditure data is robust to the presence of markups. Our approach is also highly complementary to the work by Bridgman (2016) and Bridgman (2023), which uses value-added data to estimate home productivity. Despite using quite a different empirical methodology, this work reaches the broadly consistent conclusion that the expenditure share of housework has fallen over time with rising female labor force participation.

This paper is organized as follows. Section 1 lays out the model environment. Section 2 characterizes the equilibrium, and Section 3 provides the key theoretical results, which concern the comparative statics of time allocation with respect to the cost of home inputs and the gender wage gap. Section 4 proves identification of the key model parameters, explains the data used for model estimation, and provides the parameter estimates. Section 5 describes the results of our quantitative experiments. Section 6 concludes and discusses directions for future research.

1 Model

We study a static model with heterogeneous households that derive utility from consumption of market and non-market goods. Non-market goods can be produced using a combination of home time as well as intermediate inputs, which we term *home production substitutes*. Both market goods and intermediate inputs into non-market goods are produced using labor. Each household chooses how to allocate its time between labor and home production, and how to allocate its income between market goods and intermediate inputs. The key departure relative to the existing family economics literature is that both the market good and the intermediate input for home production are produced by monopolistic firms à la Dixit and Stiglitz (1977) that charge endogenous markups.

1.1 Setup

Environment. The economy is static and populated by $G \geq 1$ types of households, indexed by $g = 1, \ldots, G$, with population shares $n_g \geq 0$ ($\sum_g n_g = 1$). Each household consists of a husband and a wife and maximizes joint utility. Types have identical preferences and home production technology but differ in market productivity, as described below. Having multiple household types is not important for the main theoretical insights, such as comparative statics results, but will be used in the model's identification. The economy has two types of firms: a continuum of firms producing differentiated market goods and a continuum of firms producing differentiated intermediate inputs into home production.

Preferences. Each household's utility is given by

$$\left[\alpha\, C^{\frac{\rho-1}{\rho}} + (1-\alpha)\, X^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}, \tag{1}$$

where $\alpha \in (0,1)$, C denotes consumption of market goods (taken to be the numéraire), X denotes consumption of non-market goods, and $\rho > 0$ is the elasticity of substitution between market goods and home-produced goods. Total consumption of market goods is a CES aggregator

$$C = \left(\int_0^1 c_j^{\frac{\eta-1}{\eta}}\, dj\right)^{\frac{\eta}{\eta-1}} \tag{2}$$

where $\eta > 1$ is the elasticity of substitution across varieties j of market goods. Similarly, total consumption of non-market goods is a CES aggregator

$$X = \left(\int_0^1 x_i^{\frac{\sigma-1}{\sigma}}\, di\right)^{\frac{\sigma}{\sigma-1}} \tag{3}$$

where σ is the elasticity of substitution across varieties i of non-market goods.

Technology. Each household is endowed with one unit of male time and one unit of female time. We assume that the husband works in the market full-time, whereas the wife's time is split between working in the market and home production.[3] All household types are equally productive at home production, but they differ in their market productivity. A household of type g supplies λ^g efficiency units of labor per unit of husband's time and ω^g efficiency units of labor per unit of wife's time worked in the market; without loss of generality, we normalize the average male efficiency units of labor $\sum_g n_g \lambda_g = 1$.[4] We denote the productivity gap within household type g by $\gamma^g := \omega^g / \lambda^g$. When analyzing a single representative household ($G = 1$), we write λ, ω, and $\gamma = \omega/\lambda$ without the superscript g.

Each variety j of market goods is produced using labor with a linear technology that converts one efficiency unit of labor into one unit of market goods. Each variety i of non-market goods can be produced using a combination of household time ℓ_i and intermediate inputs d_i according to

$$x_i = \left(\phi \ell_i^{\frac{\epsilon-1}{\epsilon}} + (1-\phi) d_i^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \tag{4}$$

where $\phi \in (0,1)$ and we assume $\epsilon, \sigma > 1$. Intermediate inputs are produced using labor with a linear technology that converts one efficiency unit of labor into $1/\zeta$ units of each intermediate input.

Market Structure. The market structure for both types of goods is monopolistically competitive à la Dixit and Stiglitz (1977). There is a monopolist firm producing each market good variety j. Similarly, there is a monopolist firm producing each intermediate input variety i. We assume that all households own both types of firms, and that firm profits are rebated to households lump-sum, with each household type receiving the profits generated by its own expenditure (an own-group rebate).[5] Labor markets are competitive.

Discussion. The model features three elasticities of substitution that govern distinct margins of adjustment. The parameter ρ governs the *between-goods* margin: how easily the household substitutes between market consumption and home-produced goods as their relative price changes. The parameter ϵ governs the *within-home-production* margin: how easily the household substitutes between its own time and purchased inputs for a given level of

[3] As documented in, e.g., Aguiar and Hurst (2007), women spend substantially more time on home production than men. Hancock, Lafortune, and Low (2025) document that this is true even in couples where women are breadwinners, that male home production time is inelastic with respect to wages, and that household chores, such as food preparation and cleaning, which we focus on in this paper, account for these patterns.

[4] This model formulation attributes gender wage gaps to differences in productivity as in Greenwood, Seshadri, and Yorukoglu (2005), but it would be straightforward to incorporate taste-based discrimination by employers into this framework, with most results unchanged.

[5] The own-group rebate isolates the deadweight loss of a markup from the cross-type redistribution of profits, so the welfare effect of removing a markup measures its efficiency cost rather than a transfer across types. Section 5 reports robustness to alternative conventions.

home production. This can be thought of as a technological parameter (see e.g. McGrattan, Rogerson, and Wright 1997 or Fang, Hannusch, and Silos 2020). The parameter σ governs the *across-variety* margin: the degree of substitutability between different varieties of home inputs, reflecting product differentiation, brand preferences, or switching costs. Home inputs are imperfect substitutes for home production time ($\epsilon < \infty$) and for each other ($\sigma < \infty$). Like Albrecht, Phelan, and Pretnar (2025) (and unlike, e.g., Atkeson and Burstein 2008 or De Loecker, Eeckhout, and Mongey 2021) the model does not assume a ranking between ϵ and σ a priori. To give an example, this implies that two different brands of frozen foods may be more easily substitutable for each other than either one is for cooking at home. On the other hand, commuting costs can make two daycare centers in different geographic regions very poor substitutes with one another, relative to either provider's substitutability with home-provided childcare. Since our stylized model is intended to capture a range of non-market goods and home activities, we allow for any configuration of these elasticities, and their magnitudes are ultimately an empirical question which we take to the data.

1.2 Household

Each household type g takes as given the prices $\{p_j\}_{j=0}^1$ of market goods, the prices $\{r_i\}_{i=0}^1$ of intermediate inputs, and the wage per efficiency unit of labor $\bar{w}$, as well as non-labor income Π^g. It maximizes (1) subject to Equations (2) to (4), as well as a budget constraint,

$$\int_0^1 p_j c_j^g \mathrm{d}j + \int_0^1 r_i d_i^g \mathrm{d}i = w_m^g + w_f^g \left(1 - \int_0^1 \ell_i^g \mathrm{d}i\right) + \Pi^g \tag{5}$$

and feasibility constraints on hours worked,

$$\int_0^1 \ell_i^g \mathrm{d}i \leq 1, \tag{6}$$

and

$$\ell_i^g \geq 0 \ \forall i, \tag{7}$$

where $w_f^g = \omega^g \bar{w}$ and $w_m^g = \lambda^g \bar{w}$ denote the female and male wage, respectively, for a type-g household; p_j denotes the price of the market-good variety j; r_i denotes the price of the intermediate input i; and $\Pi^g := \int_0^1 (r_i - \zeta\bar{w})\, d_i^g \,\mathrm{d}i + \int_0^1 (p_j - \bar{w})\, c_j^g \,\mathrm{d}j$ is the firm profit generated by type g's own purchases and rebated to it lump-sum (at $G = 1$, the economy's entire profit). The left-hand side of Equation (5) consists of the household's expenditures on market goods and intermediate inputs into home production. The right-hand side captures the household's income, noting that the female time spent working equals the time left over from home production. Equation (6) is a feasibility constraint on the total time spent in home production, where we normalize the endowment of time to 1. Equation (7) states that time in each home production activity cannot be negative. Since all types share the same preferences and face the same prices, the household problem has the same structure for each type g, differing only through the wages (w_m^g, w_f^g). The solution gives type-specific policy functions

$$\{x_i^g(\mathbf{p}, \mathbf{r}, w_m^g, w_f^g, \Pi^g),\ d_i^g(\mathbf{p}, \mathbf{r}, w_m^g, w_f^g, \Pi^g),\ \ell_i^g(\mathbf{p}, \mathbf{r}, w_m^g, w_f^g, \Pi^g),\ c_j^g(\mathbf{p}, \mathbf{r}, w_m^g, w_f^g, \Pi^g)\},$$

where $\mathbf{p} = \{p_j\}_{j\in[0,1]}$ is the vector of prices for market goods and $\mathbf{r} = \{r_i\}_{i\in[0,1]}$ is the vector of prices for home inputs.

1.3 Firms

Each market goods firm faces aggregate demand from all household types. Market good producer j chooses the price of its variety p_j given prices of its competitors $\mathbf{p_{-j}}$, prices of the home substitute goods $\mathbf{r}$, and wages $\{w_m^g, w_f^g\}_{g=1}^G$ to maximize profits

$$\pi_j^c := \max_{p_j}(p_j - \bar{w}) \sum_{g=1}^{G} n_g\, c_j^g(p_j; \mathbf{p_{-j}}, \mathbf{r}, w_m^g, w_f^g, \Pi^g) \tag{8}$$

where $c_j^g(\cdot)$ is the demand function of type g for good j taken from the household problem, and $\bar{w}$ is the marginal cost of production (the wage per efficiency unit of labor). To conserve notation, we will suppress dependence on the aggregate price vectors and profits and write simply $c_j^g(p_j)$. Similarly, home good producer i chooses the price of its variety r_i to maximize its profits

$$\pi_i^d := \max_{r_i}(r_i - \zeta\bar{w}) \sum_{g=1}^{G} n_g\, d_i^g(r_i; \mathbf{p}, \mathbf{r_{-i}}, w_m^g, w_f^g, \Pi^g) \tag{9}$$

where, again, we write $d_i^g(r_i)$ from now on.

1.4 Market clearing

The market clearing condition for this economy is the aggregate resource constraint,

$$\sum_{g=1}^{G} n_g \left[\lambda^g + \omega^g \left(1 - \int_0^1 \ell_i^g \mathrm{d}i\right)\right] = \sum_{g=1}^{G} n_g \left[\int_0^1 c_j^g \mathrm{d}j + \zeta \int_0^1 d_i^g \mathrm{d}i\right], \tag{10}$$

which states that total labor supply (in efficiency units) must equal total labor demand. When $G = 1$, this reduces to $\lambda + \gamma(1 - \int_0^1 \ell_i \mathrm{d}i) = \int_0^1 c_j \mathrm{d}j + \zeta \int_0^1 d_i \mathrm{d}i$.

1.5 Equilibrium

We are now ready to define an equilibrium of this economy:

Definition 1. *An equilibrium is a list of type-specific policy functions* $\{x_i^g(\cdot), d_i^g(\cdot), \ell_i^g(\cdot), c_j^g(\cdot)\}_{g=1}^G$, *a vector of prices* $\{\mathbf{p}, \mathbf{r}\}$, *wages* $\{w_m^g, w_f^g\}_{g=1}^G$, *and the type-specific rebated profits* $\{\Pi^g\}_{g=1}^G$ *such that*

1. *each type g's policies maximize (1) subject to Equations (2) to (7),*
2. *the prices* p_j *and* r_i *and the equilibrium profits* $\{\pi_j^c\}$ *and* $\{\pi_i^d\}$ *are solutions to Equations (8) and (9), respectively, subject to the aggregate demand functions* $\{\sum_g n_g\, c_j^g(\cdot),\ \sum_g n_g\, d_i^g(\cdot)\}$,
3. *the wages* $\bar{w}$ *and* $\{w_m^g, w_f^g\}_{g=1}^G$ *are pinned down in a competitive labor market so that* $w_m^g = \lambda^g \bar{w}$ *and* $w_f^g = \omega^g \bar{w}$,
4. *and the market clearing condition (10) is satisfied.*

2 Equilibrium characterization

2.1 Household decisions

To characterize labor supply and demand for home inputs, we solve each type's household problem taking prices as given. Since all types share the same preferences and face the same product prices, the structure of the solution is identical across types, differing only through the wages (w_m^g, w_f^g). We state the result for a generic household with wages (w_m, w_f); the type-specific demands are obtained by evaluating at $(w_m, w_f) = (w_m^g, w_f^g)$.

First, we note that constraint (7) will not bind because of Inada conditions on the functional forms. Constraint (6) binds if the wife chooses not to work, and is slack otherwise. We focus throughout on parameter values such that the equilibrium is interior, i.e. (6) does not bind and the wife's labor hours are strictly positive.[6] We can then characterize the household's optimal choice as follows:

[6] Since the model is calibrated to an economy with positive female labor force participation, this will be the empirically relevant case. A necessary and sufficient condition on parameters for an interior equilibrium, as well as the characterization of equilibrium with non-participation, are available on request.

Proposition 1. *In an interior equilibrium, the optimal choices of a household with wages w_m, w_f satisfy*

$$c_j = S_C \frac{p_j^{-\eta}}{P^{1-\eta}} I \tag{11}$$

$$x_i = S_X \frac{q_i^{-\sigma}}{Q^{1-\sigma}} I \tag{12}$$

$$d_i = (1-\phi)^{\epsilon} S_X \frac{r_i^{-\epsilon} q_i^{\epsilon-\sigma}}{Q^{1-\sigma}} I \tag{13}$$

$$\ell_i = \phi^{\epsilon} S_X \frac{w_f^{-\epsilon} q_i^{\epsilon-\sigma}}{Q^{1-\sigma}} I \tag{14}$$

where $I := w_m + w_f + \Pi$ is full income, the CES expenditure shares are

$$S_C := \frac{\alpha^{\rho} P^{1-\rho}}{\alpha^{\rho} P^{1-\rho} + (1-\alpha)^{\rho} Q^{1-\rho}}, \qquad S_X := \frac{(1-\alpha)^{\rho} Q^{1-\rho}}{\alpha^{\rho} P^{1-\rho} + (1-\alpha)^{\rho} Q^{1-\rho}} \tag{15}$$

with $S_C + S_X = 1$, and the price indices q_i, Q, and P are

$$q_i := \left[\phi^{\epsilon} w_f^{1-\epsilon} + (1-\phi)^{\epsilon} r_i^{1-\epsilon}\right]^{\frac{1}{1-\epsilon}} \tag{16}$$

$$Q := \left(\int_0^1 q_i^{1-\sigma} \mathrm{d}i\right)^{\frac{1}{1-\sigma}} \tag{17}$$

$$P := \left(\int_0^1 p_j^{1-\eta} \mathrm{d}j\right)^{\frac{1}{1-\eta}} \tag{18}$$

Proof. See Appendix A.1. □

We denote the type-g demands by $c_j^g, x_i^g, d_i^g, \ell_i^g$ and the type-specific price indices by q_i^g and Q^g, where q_i^g evaluates (16) at $w_f = w_f^g$. The market-good price index P is common across types.

The household's problem proceeds in two stages. First, the CES upper level governs the allocation of expenditure between market goods and home production: the shares S_C and S_X depend on the relative price Q/P and the elasticity ρ. When $\rho > 1$, market goods and home-produced goods are substitutes, and a rise in the home production price Q shifts expenditure toward market goods. When $\rho < 1$, they are complements. Second, given expenditure on home production $QX = S_X I$, the household cost-minimizes within each variety i by choosing the mix of time ℓ_i and purchased inputs d_i. The price index q_i captures this minimal cost.

A key feature of this two-stage structure is that the *within-home-production* decisions (the cost shares of time and inputs, and thus the price index q_i) depend only on input prices

(w_f, r_i) and the technology parameters (ϵ, ϕ), not on the upper-level preference parameters (α, ρ). This separation is central to our identification strategy.

2.2 Firm pricing decisions

Market Goods Producers. Firm j chooses p_j to maximize the objective in Equation (8). Since each type's demand $c_j^g(p_j)$ has the constant-elasticity form given by Equation (11), the aggregate demand $\sum_g n_g c_j^g(p_j)$ inherits the same elasticity η with respect to p_j. Standard derivations then yield that the price p_j is a markup of the marginal cost $\bar{w}$,

$$p_j = \frac{\eta}{\eta - 1}\bar{w} \tag{19}$$

where $\eta/(\eta-1)$ is the standard constant-elasticity markup over marginal cost; equivalently, the Lerner index (the price–cost margin) is $(p_j - \bar{w})/p_j = 1/\eta$ (Lerner 1934). In particular, the market good markup does not depend on the number of types or the distribution of wages.

Home Goods Producers. Firm i chooses r_i to maximize the objective in Equation (9). We first characterize the solution in the representative-agent case ($G = 1$), where the firm faces demand $d_i(r_i)$ from a single household with female wage w_f. Substituting Equation (13) into the firm's objective, we get

$$\pi_i^d = \max_{r_i}(r_i - \zeta\bar{w})(1-\phi)^{\epsilon} S_X \frac{r_i^{-\epsilon} q_i^{\epsilon-\sigma}}{Q^{1-\sigma}} I \tag{20}$$

We obtain the following characterization of the solution:

Proposition 2. *When $G = 1$, the firm's problem in Equation (20) has a unique solution r_i, which satisfies the equation*

$$\frac{r_i}{r_i - \zeta\bar{w}} = \frac{\phi^{\epsilon} w_f^{1-\epsilon}}{\phi^{\epsilon} w_f^{1-\epsilon} + (1-\phi)^{\epsilon} r_i^{1-\epsilon}} \cdot \epsilon + \frac{(1-\phi)^{\epsilon} r_i^{1-\epsilon}}{\phi^{\epsilon} w_f^{1-\epsilon} + (1-\phi)^{\epsilon} r_i^{1-\epsilon}} \cdot \sigma \tag{21}$$

Proof. See Appendix A.2. □

Multi-type markup. When $G > 1$, each home good firm faces aggregate demand $D_i = \sum_{g=1}^{G} n_g\, d_i^g(r_i)$, where the type-specific demands d_i^g depend on the type-specific price index q_i^g, which involves w_f^g and therefore differs across types. The elasticity $\xi_i = -\partial \ln D_i/\partial \ln r_i$ of aggregate demand with respect to r_i is the demand-share-weighted average of the type-specific elasticities: $\xi_i = \sum_{g=1}^{G} \frac{n_g\, d_i^g}{D_i} \cdot \xi_i^g$, where ξ_i^g is the demand elasticity perceived by the firm from type g, which has the same form as the left-hand side of (21) evaluated at $w_f = w_f^g$. The firm's optimal price then satisfies

$$\frac{r_i}{r_i - \zeta \bar{w}} = \xi_i = \sum_{g=1}^{G} \frac{n_g\, d_i^g}{D_i} \left\{ \frac{\phi^\epsilon (w_f^g)^{1-\epsilon}}{\phi^\epsilon (w_f^g)^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}} \cdot \epsilon + \frac{(1-\phi)^\epsilon r_i^{1-\epsilon}}{\phi^\epsilon (w_f^g)^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}} \cdot \sigma \right\} \tag{22}$$

generalizing Proposition 2. The representative-agent case ($G = 1$) is recovered when there is a single type with $w_f^1 = w_f$.

2.3 Equilibrium allocations

Within each type, symmetry across varieties implies $c_j^g = C^g$ for all j and $x_i^g = X^g$, $d_i^g = d^g$, and $\ell_i^g = \ell^g$ for all i. Similarly, we can impose $p_j = P$, $r_i = r$, and $q_i^g = Q^g$ for all i, j. Normalizing $P = 1$, the wage per efficiency unit $\bar{w}$ is determined from Equation (19) as

$$\bar{w} = \frac{\eta - 1}{\eta}. \tag{23}$$

It will be convenient to define the markup on home inputs,

$$\mu := \frac{r}{\zeta \bar{w}} \tag{24}$$

We now characterize the equilibrium allocations for the representative-agent case ($G = 1$), which we use for the comparative statics in Section 3. We normalize the husband's market productivity $\lambda = 1$, so that $\gamma = \omega$. The equilibrium male wage is then $w_m = \bar{w}$, which is given by Equation (23), and the equilibrium female wage is $w_f = \gamma \bar{w}$, giving

$$w_f = \gamma \frac{\eta - 1}{\eta}. \tag{25}$$

Given w_m and w_f, we can solve for the equilibrium price r from Equation (21), which in turn determines the equilibrium Q via Equation (16). The markup μ is then pinned down from Equation (21) as

$$\frac{\mu}{\mu - 1} = \frac{\phi^\epsilon \gamma^{1-\epsilon}}{\phi^\epsilon \gamma^{1-\epsilon} + (1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}} \cdot \epsilon + \frac{(1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}}{\phi^\epsilon \gamma^{1-\epsilon} + (1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}} \cdot \sigma \tag{26}$$

The equilibrium price Q follows from Equation (16) as

$$Q = \frac{\eta - 1}{\eta} [\phi^\epsilon \gamma^{1-\epsilon} + (1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}]^{\frac{1}{1-\epsilon}} \tag{27}$$

Turning to quantities, we introduce two objects that summarize the equilibrium structure. The *cost share of purchased inputs within home production* is

$$s := \frac{rd}{w_f \ell + rd} = \frac{(1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}}{\phi^\epsilon \gamma^{1-\epsilon} + (1-\phi)^\epsilon (\mu\zeta)^{1-\epsilon}} \tag{28}$$

which is determined entirely by cost minimization within X and depends on $(\epsilon, \phi, \zeta, \gamma, \mu)$ but not on (α, ρ). The *expenditure ratio* of home production to market goods is

$$R := \frac{QX}{C} = \left(\frac{1-\alpha}{\alpha}\right)^\rho Q^{1-\rho} \tag{29}$$

which is determined by the CES upper level and depends on (α, ρ, Q). When $\rho = 1$, the expenditure ratio $R = (1-\alpha)/\alpha$ is constant: every household devotes the same fraction of income to home production regardless of its cost. When $\rho > 1$, R falls with Q: as home production becomes more expensive, households substitute toward market goods. The equilibrium allocations are then uniquely determined by the resource constraint

$$\gamma\ell + C + \zeta d = 1 + \gamma, \tag{30}$$

together with Equations (28) and (29) and the symmetric versions of Equations (13) and (14):

Proposition 3. *In the representative-agent case ($G = 1$), define $\mathcal{D} := \bar{w} + R\big[(1-s) + s/\mu\big]$. The symmetric equilibrium allocations are*

$$C = \frac{\bar{w}_m\,(1+\gamma)}{\mathcal{D}} \tag{31}$$

$$\ell = \frac{R(1-s)}{\gamma} \cdot \frac{1+\gamma}{\mathcal{D}} \tag{32}$$

$$d = \frac{Rs}{\mu\zeta} \cdot \frac{1+\gamma}{\mathcal{D}} \tag{33}$$

$$X = \frac{R\,\bar{w}_m}{Q} \cdot \frac{1+\gamma}{\mathcal{D}} \tag{34}$$

where $\bar{w}_m = (\eta-1)/\eta$, R is given by (29), s by (28), and μ by Equation (26).

Proof. See Appendix A.3. □

When $G > 1$, Equation (22) implies that the markup μ is instead determined by

$$\frac{\mu}{\mu-1} = \sum_{g=1}^{G} \frac{n_g\, d_i^g}{D_i} \left\{ \frac{\phi^{\epsilon}(\omega^g)^{1-\epsilon}}{\phi^{\epsilon}(\omega^g)^{1-\epsilon} + (1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}} \cdot \epsilon + \frac{(1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}}{\phi^{\epsilon}(\omega^g)^{1-\epsilon} + (1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}} \cdot \sigma \right\} \tag{35}$$

The type-specific composite price index Q^g is determined by

$$Q^g = \frac{\eta-1}{\eta}[\phi^{\epsilon}(\omega^g)^{1-\epsilon} + (1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}]^{\frac{1}{1-\epsilon}} \tag{36}$$

The quantities (C^g, X^g, d^g, ℓ^g) have the functional forms given by the symmetric versions of Equation (11), Equation (12), Equation (13) and Equation (14), evaluated at the type-specific wages (w_m^g, w_f^g) with the corresponding type-specific cost share

$$s^g := \frac{rd^g}{w_f^g \ell^g + rd^g} = \frac{(1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}}{\phi^{\epsilon}(\omega^g)^{1-\epsilon} + (1-\phi)^{\epsilon}(\mu\zeta)^{1-\epsilon}} \tag{37}$$

and expenditure ratio

$$R^g := \frac{Q^g X^g}{C^g} = \left(\frac{1-\alpha}{\alpha}\right)^{\rho} (Q^g)^{1-\rho} \tag{38}$$

Together with the market-clearing condition (10), this system fully describes the equilibrium in the multi-type case. We focus on the $G = 1$ case below for the analytical results, and return to the multi-type equilibrium in the estimation and quantitative analysis.

Discussion: model mechanisms. The equilibrium ℓ in (32) makes transparent the two channels through which the wife's wage affects home production time. The cost share $(1 - s)$ captures the *within-home-production* channel: a higher γ raises w_f, making time more expensive relative to purchased inputs and lowering $(1 - s)$ for a given Q. The expenditure ratio R captures the *between-goods* channel: a higher γ raises the home production price Q, and when $\rho > 1$, the household responds by reallocating expenditure toward market goods, lowering R. Both channels reduce ℓ when the wife's wage rises.

The markup on home inputs μ creates a wedge that distorts equilibrium allocations, particularly the allocation of time between market work and home production. The key insight comes from examining equation (26). At the extremes, the formula simplifies to standard cases, which we further discuss below in Section 2.4. When $\phi = 1$ (households rely entirely on time), we get $\frac{\mu}{\mu-1} = \epsilon$, reflecting the elasticity of substitution between home time and purchased home inputs. When $\phi = 0$ (households rely entirely on purchased inputs), we obtain $\frac{\mu}{\mu-1} = \sigma$, reflecting the elasticity of substitution between non-market good varieties.

More generally, $\frac{\mu}{\mu-1}$ represents a weighted average of these two elasticities, where the weights are endogenously determined by the parameters ϕ, ζ, and γ. When households are highly reliant on their own time for provision of non-market goods – either because home time is very productive (ϕ is high), intermediate inputs require substantial labor to produce (ζ is high), or female wages are relatively low (γ is low) – each intermediate input firm behaves as if household time is its key competitor. In this case, the relevant elasticity is ϵ, and $\frac{\mu}{\mu-1}$ approaches this value. Conversely, when households are not very reliant on their home time and rely heavily on purchased home inputs, intermediate input firms face competition primarily from other input varieties rather than from home time. The relevant elasticity becomes σ, and $\frac{\mu}{\mu-1}$ approaches this value accordingly.

An implication of this analysis is that an increase in the reliance on home inputs relative to household time may raise or lower the markup on home inputs. We confirm this in the comparative statics results of Section 3.

2.4 Nested cases

Our model combines monopolistic competition with a relatively standard framework from family economics. In particular, the model nests several special cases prominent in the literature. When $\rho \to 1$, the CES upper level reduces to Cobb-Douglas preferences $\alpha \ln C + (1-\alpha) \ln X$, and the expenditure ratio $R = (1-\alpha)/\alpha$ is constant across household types. This shuts down the between-goods substitution channel. When $\phi = 1$, home-produced goods use time only, and the model then reduces to the standard Dixit and Stiglitz (1977) monopolistic competition framework with endogenous labor supply. In that case, the markup on home inputs is irrelevant and the markup on market goods acts as a conventional distortionary tax.

At the other extreme, when $\phi = 0$, the model reduces to a Dixit and Stiglitz (1977) monopolistic competition framework with two classes of goods with different elasticities of substitution, which determine each product's respective markup. In this case, the markup on market goods is equal to $\frac{\eta}{\eta-1}$ and the markup on home goods is equal to $\mu = \frac{\sigma}{\sigma-1}$.

Next, if the elasticity of substitution in either sector is taken to infinity, the respective product market approaches perfect competition. Taking the limits $\eta \to \infty$ and $\sigma \to \infty$ recovers the usual framework in family macroeconomics (Greenwood, Guner, and Vandenbroucke 2017). In this case, the markup on home inputs is $\mu = 1$, and so their price equals marginal cost, $r = \zeta \bar{w}$.[7] Our framework therefore permits very straightforward comparisons to well-known benchmarks, which we will elaborate on below in the quantitative analysis.

[7] It is also straightforward to show that Pareto efficiency requires $\eta \to \infty$ and $\sigma \to \infty$; as is standard in the Dixit and Stiglitz (1977) literature, the markup in each sector acts as a distortionary tax.

3 Comparative statics

In this section, we study how market power for home production substitutes distorts labor supply and interacts with other outcomes in the model. Our analytical results focus on the the representative-agent case ($G = 1$), for which the equilibrium allocation is characterized in Proposition 3.

Direct effects of market power. We first confirm the intuitive result that market power in *either* market distorts labor supply.

Lemma 1. *$\frac{d\ell}{d\eta} < 0$ and $\frac{d\ell}{d\sigma} < 0$.*

Proof. This follows directly from differentiating Equation (32) with respect to each parameter. □

As is well known, market power in the market goods sector distorts labor supply, acting as a distortionary tax on market consumption. In addition, market power for home production substitutes creates a novel, analogous distortion: by inducing substitution away from purchased home inputs and instead towards home production time, it induces households to substitute toward home production and away from market work.

We next revisit classic questions in family macroeconomics, namely how time allocation, captured here by the female time in home production ℓ, depends on the narrowing gender wage gap, captured by an increase in γ, and technological progress in home substitutes, captured by a fall in ζ. The key insight here is that market power in the home input market interacts with these economic forces in a way that market power in the market goods sector does not. Moreover, the endogenous response of markups can either dampen or amplify the effects of other economic trends, depending on the relative substitutability of time and household inputs.

Technological progress in home inputs. We first analyze the effect of a decline in the cost of producing home inputs due to technical change. Differentiating Equation (26) with respect to ζ, we get the following result:

Lemma 2. *The comparative statics properties of the equilibrium markup with respect to ζ are as follows:*

1. *If $\epsilon > \sigma$, then $-\frac{\mu}{\zeta} < \frac{d\mu}{d\zeta} < 0$.*

2. *If $\epsilon = \sigma$, then $\frac{d\mu}{d\zeta} = 0$.*

3. *If* $\epsilon < \sigma$, *then* $\frac{d\mu}{d\zeta} > 0$.

Proof. See Appendix A.4. $\square$

A decrease in the marginal cost ζ of producing home inputs makes households more reliant on home inputs and less reliant on home time in equilibrium. If $\epsilon > \sigma$, this means that households are more reliant on a less easily substitutable product. Producers of home inputs then face a lower elasticity of demand and optimally raise their markups. Lemma 2 establishes that prices still fall on net, but the fall is dampened by the rise in markups. If $\epsilon < \sigma$, home inputs are very easily substitutable with one another. The increased reliance on home inputs then raises the elasticity of demand and lowers markups. In this case, the fall in the price is amplified by the endogenous response of markups.

Next, we examine the implications for time allocation. We can decompose

$$\frac{d\ell}{d\zeta} = \frac{\partial \ell}{\partial \zeta} + \frac{\partial \ell}{\partial \mu}\frac{d\mu}{d\zeta},$$

where $\frac{d\ell}{d\zeta}$ is the total derivative of ℓ with respect to ζ, $\frac{\partial \ell}{\partial \zeta}$ is the direct effect of ζ keeping μ fixed, and $\frac{\partial \ell}{\partial \mu}\frac{d\mu}{d\zeta}$ is the effect operating through the equilibrium markup. It is straightforward to verify from Equation (32) that $\frac{\partial \ell}{\partial \zeta} > 0$, i.e., the direct effect of a decrease in the cost of home inputs is to lower home production time and raise labor supply. This is the standard result in family economics emphasized, e.g., in Greenwood, Seshadri, and Yorukoglu (2005). The novelty is that the fall in ζ is not perfectly passed through to prices, and as shown in Lemma 2, the response of the markup can be either dampening or amplifying. It is straightforward to verify that $\frac{\partial \ell}{\partial \mu} > 0$, and so the direction of the equilibrium effect depends on the sign of $\frac{d\mu}{d\zeta}$. We summarize this result in the following proposition:

Proposition 4. *We have* $\frac{d\ell}{d\zeta} > 0$ *and*

1. $\frac{d\ell}{d\zeta} < \frac{\partial \ell}{\partial \zeta}$ *if* $\epsilon > \sigma$, *or*
2. $\frac{d\ell}{d\zeta} = \frac{\partial \ell}{\partial \zeta}$ *if* $\epsilon = \sigma$, *or*
3. $\frac{d\ell}{d\zeta} > \frac{\partial \ell}{\partial \zeta}$ *if* $\epsilon < \sigma$.

As stressed in the family economics literature, a falling cost of home inputs has the effect of raising female labor supply. Here, the cost is not passed through to prices one-for-one, and the rise in labor supply may be either dampened or amplified by the endogenous markup response, following the logic of Lemma 2.

Narrowing the gender wage gap. We similarly consider the effect of a smaller gender wage gap, captured here by an increase in γ. Differentiating Equation (26) with respect to γ, we get the result:

Lemma 3. *The comparative statics properties of the equilibrium markup with respect to γ are as follows:*

1. *If $\epsilon > \sigma$, then $0 < \frac{d\mu}{d\gamma} < \frac{\mu}{\gamma}$.*
2. *If $\epsilon = \sigma$, then $\frac{d\mu}{d\gamma} = 0$.*
3. *If $\epsilon < \sigma$, then $\frac{d\mu}{d\gamma} < 0$.*

Proof. See Appendix A.4. □

The logic of Lemma 3 parallels that of Lemma 2. A higher γ raises the opportunity cost of home production, making households relatively more reliant on home inputs. The effect on markups depends on whether home inputs are highly substitutable with each other. Similarly to the above discussion of technological progress, we use this result to examine comparative statics of ℓ with respect to γ. The overall effect of a rise in γ on ℓ can be decomposed into the direct effect $\frac{\partial \ell}{\partial \gamma}$ and the equilibrium effect through μ:

$$\frac{d\ell}{d\gamma} = \frac{\partial \ell}{\partial \gamma} + \frac{\partial \ell}{\partial \mu}\frac{d\mu}{d\gamma},$$

The direct effect $\frac{\partial \ell}{\partial \gamma}$ is negative, as can be verified from Equation (32). Intuitively, this result follows for two reasons. First, a higher γ raises the opportunity cost of home production, which lowers ℓ. Second, a higher γ raises household income, which further lowers ℓ since a richer household can purchase more home inputs in the market. The direct effect is thus necessarily to reduce time spent in home production. The equilibrium effect through μ can be dampening or amplifying, similarly to the discussion above. We summarize this result in the following proposition:

Proposition 5. *We have $\frac{d\ell}{d\gamma} < 0$ and*

1. *$\frac{d\ell}{d\gamma} > \frac{\partial \ell}{\partial \gamma}$ if $\epsilon > \sigma$;*
2. *$\frac{d\ell}{d\gamma} = \frac{\partial \ell}{\partial \gamma}$ if $\epsilon = \sigma$;*
3. *$\frac{d\ell}{d\gamma} < \frac{\partial \ell}{\partial \gamma}$ if $\epsilon < \sigma$.*

This simple result has potentially non-trivial policy implications. Policies that reduce the gender wage gap may be more or less effective, in a quantitative sense, depending on the market power in the home inputs market. Intuitively, a policy reducing the gender wage gap would, all else equal, draw more women into the labor force; but this effect may be offset by endogenously rising prices of home production substitutes, which in turn make it more difficult to enter the labor force despite higher wages.

The role of the between-goods elasticity ρ. The decompositions above isolate the markup channel by holding μ fixed in the "direct effect." With CES upper-level preferences, this direct effect incorporates not only the within-home-production cost adjustment (governed by ϵ) but also the reallocation of expenditure between market goods and home production (governed by ρ). When $\rho > 1$, a fall in ζ or a rise in γ reduces the home production price Q, causing the household to shift expenditure *toward* home production, partially offsetting the within-X substitution away from time. When $\rho < 1$, the expenditure channel reinforces the within-X channel. Importantly, the markup channel (the comparison between $d\ell/d\zeta$ and $\partial\ell/\partial\zeta$) is governed entirely by the within-X parameters ϵ and σ through the markup equation (26), which does not depend on ρ. The upper-level elasticity ρ therefore affects the *magnitude* of the direct effect but not the *sign* of the markup amplification or dampening.

4 Estimation

The model has seven structural parameters $\{\alpha, \rho, \eta, \sigma, \epsilon, \phi, \zeta\}$. We first establish a sufficient condition for all parameters to be identified, and then describe the data used to implement the estimation.

4.1 Identification

The estimation uses data on type-specific wages, time use, and expenditure patterns. We begin by defining the normalized wages that appear throughout the identification argument. Since we normalize $\sum_g n_g \lambda_g = 1$, the population-average male wage $\bar{w}_m$ directly corresponds to the wage per efficiency units of labor, $\bar{w}$, in the model. Wage w_f^g and w_m^g for each type then yield the type-specific productivities $\omega^g := w_f^g / \bar{w}_m$ and $\lambda^g := w_m^g / \bar{w}_m$. Time use and expenditure patterns are then used to construct the empirical cost shares s^g and R^g.

The identification strategy exploits the model's two-stage structure. Each type's cost share s^g within home production (evaluated from Equation (28) at the respective ω^g) depends only on input prices and (ϵ, ϕ), not on the upper-level parameters (α, ρ). The expenditure ratio $R^g = QX^g/PC^g$ depends on the home production price index Q^g and (α, ρ). This separation allows us to identify the within-home-production parameters from variation in s^g across types, and the between-goods parameters from variation in R^g.

Theorem 1 (Constructive identification). *Suppose we observe the market good markup μ_M, the home substitute markup μ, the ratios of prices to population-average male wages $r/\bar{w}_m$ and $P/\bar{w}_m$, and type-specific moments $\{\omega^g, \ell^g, s^g, R^g\}_{g=1}^G$ for $G \geq 2$ household types with $\omega^g \neq \omega^{g'}$ for some $g \neq g'$. Then all seven structural parameters $\{\eta, \zeta, \epsilon, \phi, \rho, \alpha, \sigma\}$ are uniquely determined and can be recovered constructively:*

1. *η from μ_M;*
2. *ζ from $r/(\mu\bar{w}_m)$;*
3. *ϵ from the cross-group gradient in s^g;*
4. *ϕ from the level of s^g;*
5. *ρ from the cross-group gradient in R^g;*
6. *α from the level of R^g;*
7. *σ from the aggregate markup equation.*

Proof. Step 1 (η). From (19), the market good price satisfies $P = \frac{\eta}{\eta-1}\bar{w}_m$, so the markup is $\mu_M = \frac{\eta}{\eta-1}$ and $\eta = \mu_M/(\mu_M - 1)$.

Step 2 (ζ). From (24), the equilibrium price of home inputs is $r = \mu\zeta\bar{w}_m$, which gives $\zeta = r/(\mu\bar{w}_m)$.

Step 3 (ϵ). The cost share s^g in (28) depends on $(\epsilon, \phi, \zeta, \omega^g, \mu)$ but not on (α, ρ), since it is determined by cost minimization within X. Evaluating (28) at type-specific wages and taking log-odds yields

$$\ln \frac{1 - s^g}{s^g} = \epsilon \ln \frac{\phi}{1 - \phi} + (1 - \epsilon) \ln \frac{\omega^g}{\mu\zeta}$$

The ratio across groups g and g' cancels (ϕ, ζ, μ) and yields

$$\epsilon = 1 - \frac{\ln\left[(1 - s^{g'})s^g \,/\, (s^{g'}(1 - s^g))\right]}{\ln(\omega^{g'}/\omega^g)} \tag{39}$$

which is well-defined whenever $\omega^g \neq \omega^{g'}$.

Step 4 (ϕ). Given (ϵ, ζ, μ) from Steps 1–3, ϕ is recovered from the level of s^g by rearranging (28),

$$\frac{\phi}{1-\phi} = \left[\frac{1-s^g}{s^g} \cdot \left(\frac{\mu\zeta}{\omega^g}\right)^{1-\epsilon}\right]^{1/\epsilon}$$

Step 5 (ρ). Taking logs of (29) gives $\ln R^g = \rho \ln\left(\frac{1-\alpha}{\alpha}\right) + (1-\rho)\ln Q^g$, where Q^g follows from (27) evaluated at type-specific wages, $Q^g = \bar{w}_m\left[\phi^\epsilon(\omega^g)^{1-\epsilon} + (1-\phi)^\epsilon(\mu\zeta)^{1-\epsilon}\right]^{1/(1-\epsilon)}$, and is computable from Steps 1–4. Differencing across groups ($\bar{w}_m$ cancels) gives

$$\rho = 1 - \frac{\ln(R^{g'}/R^g)}{\ln(Q^{g'}/Q^g)} \tag{40}$$

Step 6 (α). Given ρ, the level of R^g in (29) pins down α via

$$\frac{1-\alpha}{\alpha} = \left[R^g \cdot (Q^g)^{\rho-1}\right]^{1/\rho}$$

Step 7 (σ). The aggregate markup condition (35) takes the form $\mu/(\mu-1) = \bar{w}_1\epsilon + \bar{w}_2\sigma$, where $\bar{w}_1$ and $\bar{w}_2$ are demand-share-weighted averages of the type-specific markup weights from Proposition 2 and the demand shares $n_g d^g/D$ are computed from observed expenditure (r cancels in $n_g r d^g / \sum_h n_h r d^h$). This is linear in σ, so

$$\sigma = \frac{\mu/(\mu-1) - \bar{w}_1\epsilon}{\bar{w}_2}$$

All seven parameters are thus recovered in closed form as functions of the data, completing the proof. □

Discussion. The identification cleanly separates two substitution margins. The *within-home-production* margin (ϵ) is identified from the cost share s^g, which depends only on input prices and (ϵ, ϕ). The *between-goods* margin (ρ) is identified from the expenditure ratio R^g, which depends on the home production price index Q^g and (α, ρ). Although Equations (39) and (40) are stated as pairwise differences, each identifies the slope of a log-linear relationship across types (that of $\ln \frac{1-s^g}{s^g}$ on $\ln \omega^g$ with slope $(1-\epsilon)$, and of $\ln R^g$ on $\ln Q^g$ with slope $(1-\rho)$). The system is therefore exactly identified at $G = 2$ and over-identified for $G \geq 3$, and we estimate each slope by a population-weighted OLS regression across the four groups.[8] The ordering $\epsilon < \sigma$, and with it every quantitative conclusion in

[8] Equivalently, the same moments can be stacked and the parameters estimated jointly by the generalized method of moments (GMM), rather than recovered sequentially as in Theorem 1. Weighting each moment equally and the four groups by population, this delivers estimates close to those we report: a within-home-production elasticity $\epsilon = 1.50$ and an across-variety elasticity $\sigma = 42.2$, against the 1.51 and 39.95 of our

Section 5, is unchanged. Importantly, Step 3 uses only cross-group *ratios* of s^g, in which the price parameters (μ, ζ) cancel, so ϵ does not depend on the NielsenIQ price data or the symmetry assumption. Steps 4–7 combine data from multiple sources and require the Dixit-Stiglitz symmetry assumption to reconcile NielsenIQ prices with CEX quantities. Finally, note that our identification relies on how the cost shares, i.e. the *ratios* of household time to input purchases, vary across household types; it does not use the variation in the *levels* of time use across household types. Indeed, our identification of ϵ uses the fact that variation in cost shares across household types reflects substitution alone; this is not true of variation in time use itself, which also reflects income effects. The variation in the levels of time use across household types is therefore not targeted in the estimation and will be used for model validation.

4.2 Data

Our estimation strategy requires $G \geq 2$ household types with distinct wages. It also requires data on each type's allocation of time and expenditures on market goods and home production inputs. Finally, the estimation also requires data on markups for market goods and home production inputs. The latter is the most important difference from recent literature estimating home production technology (e.g. Bridgman, Craig, and Kanal (2022), Bridgman (2023), Boerma and Karabarbounis (2021), Fang and Yang (2022), Fang, Hannusch, and Silos (2020)) which either takes prices as given or assumes competitive markets, and thus requires data on time allocation and expenditures only. Implementing this estimation strategy requires merging several data sources and, further, requires classification of products as market goods vs. home production substitutes to be consistent across datasets.

We combine several data sources: NielsenIQ scanner data for the markup moments (μ_M, μ), the NielsenIQ Consumer Panel for the unit price (r), the ATUS for housework time (ℓ^g), the ATUS-CPS linked file for wages (ω^g, λ^g), and the Consumer Expenditure Survey (CEX) for expenditure moments (s^g, R^g). In our baseline analysis, we identify housework (interior cleaning and laundry) as home production, motivated by the fact that this activity is unambiguously a chore, and the model abstracts from leisure, treating home production as a pure technology.

NielsenIQ Scanner data (2007). We use product-module-level retail markup estimates from Sangani (2025) to construct sales-weighted average markups for market goods and housework substitutes. Products are first classified into market goods and home substitutes

baseline (Table 2).

using a two-stage keyword-based procedure (Appendix B.2), then home substitutes are further classified into food and cleaning sub-categories. The housework markup $\mu = 1.25$ is the sales-weighted average over 46 cleaning-subcategory modules (e.g. detergents, paper towels, disinfectants, fabric softeners).

American Time Use Survey (2019–2024). The ATUS provides time diaries for a nationally representative sample. We restrict to survey years 2019–2024 (to match the CEX vintage), married women with spouse present where both spouses are employed, and measure housework time as the sum of daily minutes in interior cleaning (code 020101) and laundry (code 020102). Food and drink preparation (020201) and kitchen cleanup (020203) are excluded, consistent with the restriction to housework. The time endowment is $T = 5.34$ hours per day, computed as the population-weighted average of housework plus market work time over the wives in these dual-earner couples (the same sample used for the wage and expenditure moments).

ATUS-CPS linked file (2019–2024). Wages are constructed from the CPS earnings data linked to ATUS respondents. We restrict to survey years 2019–2024 to match the CEX vintage. We compute hourly wages and link married couples using the CPS spouse pointer. This yields the population-average male wage $\bar{w}_m = \$34.63$/hour, type-specific wages, and population shares.

NielsenIQ Consumer Panel (2019). We compute the unit price of housework substitutes $r = \$0.10$ per ounce from the 2019 wave, as total expenditure divided by total ounces purchased over the ounce-measured cleaning consumables (e.g. detergents, bleach, all-purpose and bathroom cleaners, liquid softeners), which avoids averaging dollars across incomparable package counts and sizes. Markups are retained from the 2007 vintage, the latest year for which wholesale cost data are available.

Consumer Expenditure Survey (2024). The CEX provides comprehensive household expenditure data. We pool the 2024 interview files (five quarters), aggregate to one observation per consumer unit across interviews, and restrict to married households in which both spouses are employed. Housework substitute expenditure rd^g is the sum of spending on cleaning and laundry supplies (from the CEX Diary, UCCs 330119 and 330310) and hired cleaning services (UCCs 340310, 340520, 340530 from the Interview MTBI file).[9] Market

[9] Food expenditure (both food at home and food away from home) is excluded from rd^g and remains in market good expenditure PC^g, consistent with the housework scope restriction.

Table 1: Group-Specific Data ($G = 4$)

Group	Wife	Husband	w_f^g	w_m^g	ω^g	λ^g	n^g	s^g	R^g	rd^g
1	No BA	No BA	\$21.24	\$24.88	0.613	0.719	0.348	0.069	0.230	\$483
2	No BA	BA+	\$25.32	\$38.55	0.731	1.113	0.073	0.114	0.192	\$768
3	BA+	No BA	\$35.87	\$30.22	1.036	0.873	0.183	0.074	0.253	\$662
4	BA+	BA+	\$40.05	\$44.50	1.156	1.285	0.396	0.102	0.235	\$940

Note: Wages from ATUS-CPS linked file (2019–2024), restricted to married couples where the wife is the ATUS respondent and both spouses report valid wages. Population-average male wage is $\bar{w}_m = \$34.63$/hour. Cost shares s^g and expenditure ratios R^g combine CEX 2024 housework expenditure with CPS wages and ATUS housework time (both 2019–2024). Population shares n^g from the linked-couples sample. rd^g is annual housework substitute expenditure (housekeeping supplies + cleaning services).

good expenditure PC^g is total expenditure less the hired cleaning services and a set of non-consumption and durable categories, detailed in Appendix B.2.[10] The cost share s^g and expenditure ratio R^g combine CEX expenditure with CPS wages and ATUS housework time.

4.3 Estimates

Sales-weighted average markups are $\mu_M = 1.36$ for market goods and $\mu = 1.25$ for housework substitutes. Note that this does *not* imply that the elasticities of substitution are similar across product categories. In fact, as described below, our estimation will imply a much higher value for σ than for η; the reason for similar markups is that markups for home substitutes also depend on the substitutability between time and inputs, which we will estimate to be low.

Table 1 displays the group-specific summary statistics. The four groups span a wide range of wages: the wife's normalized wage ω^g ranges from 0.61 (Group 1) to 1.16 (Group 4), providing the cross-group variation needed for identification. The cost share s^g ranges from 0.07 to 0.11, reflecting the fact that housework is a time-intensive activity where purchased inputs (cleaning supplies, hired cleaners) account for a modest share of total production cost. The expenditure ratio R^g ranges from 0.19 to 0.25, reflecting the small share of total household expenditure devoted to housework production. The expenditure ratio is approximately flat across groups, consistent with $\rho \approx 1$: as the price of housework rises with the wife's wage, households neither sharply increase nor decrease their housework spending share.

[10] We deliberately adopt a current, non-durable, non-housing consumption concept for PC^g: from total expenditure we strip personal insurance and pensions, cash contributions, and education (savings, transfers, and investment), together with all shelter, health insurance, and new and used vehicle purchases. These categories are together roughly half of total expenditure and grow steeply with income, representing durable or non-consumption outlays rather than current consumption of market goods in the model's sense.

Table 2: Parameter Estimates

Parameter	Description	Estimate	Source
α	CES weight on market goods	0.83	Level-identified
ρ	Elasticity between C and X	0.90	Estimated (4-group gradient)
η	Elasticity across market goods	3.74	$\mu_M/(\mu_M - 1)$
ϵ	Elasticity between time and inputs	1.51	Estimated (4-group gradient)
ϕ	Weight on time in home production	0.97	Level-identified
ζ	Technology parameter	0.0023	$r/(\mu\bar{w}_m)$
σ	Elasticity across home goods	39.95	Lerner condition

Note: Identification using $G = 4$ groups defined by wife's $\times$ husband's education. The within-home-production elasticity ϵ is estimated from the population-weighted cross-group gradient in housework cost shares. The between-goods elasticity ρ is estimated from a population-weighted OLS regression of $\log R^g$ on $\log Q^g$ across the four groups. The remaining parameters are level-identified from population-weighted average moments as in Steps 1, 2, 4, 6, and 7 of Theorem 1. Markups from Sangani (2025), 2007 vintage, restricted to 46 cleaning-subcategory modules; unit price from NielsenIQ Consumer Panel, 2019, cleaning modules; wages and time from ATUS-CPS, 2019–2024; expenditure from CEX, 2024.

Table 2 reports the parameter estimates. The within-home-production elasticity $\epsilon = 1.51$ is estimated from the population-weighted cross-group gradient in cost shares across the four education groups. The between-goods elasticity $\rho = 0.90$ is estimated from the cross-group gradient in expenditure ratios using a population-weighted regression of $\log R^g$ on $\log Q^g$ across the four groups, and implies that market goods and home-produced housework are approximately unit-elastic substitutes. The remaining parameters are level-identified as in Theorem 1. The weight on time $\phi = 0.97$ indicates that housework relies heavily on the wife's own time, with purchased inputs accounting for only about 9% of production costs at the population average. The technology parameter $\zeta = 0.0023$ reflects the unit cost of housework substitutes relative to wages. Most importantly, the elasticity across home good varieties $\sigma = 39.95$ is substantially larger than ϵ, implying that different brands of housework substitutes are much more substitutable with each other than any of them is with the household's own time. This result is robust to different ways of disaggregating the data. Since $\epsilon < \sigma$, the comparative statics results in Section 3 imply that endogenous markups *amplify* the direct effects of technological progress and wage convergence on female labor supply.

4.4 Model fit

The identification uses the cross-group gradient in s^g for ϵ and R^g for ρ, with the remaining parameters recovered from level moments at the population-weighted average. Table 3 evaluates the model against the data moments and the model's predicted housework time ℓ^g for all four groups. Because ϵ and ρ are identified from the full four-group gradient (rather than from two targeted groups), no individual group's moments are matched exactly by construction. The model-predicted ℓ^g is an equilibrium outcome that must be internally consistent with the predicted s^g and R^g, so the fit on ℓ^g is a cross-equation consistency check.

Table 3: Model Fit

		Group 1	Group 2	Group 3	Group 4
Cost share s^g	Data	0.069	0.114	0.074	0.102
	Model	0.071	0.077	0.091	0.096
Expenditure ratio R^g	Data	0.230	0.192	0.253	0.235
	Model	0.225	0.229	0.236	0.238
Housework time (hrs/day)	Data	0.85	0.65	0.63	0.57
	Model	0.69	0.81	0.60	0.69

Note: Housework time is interior cleaning + laundry from the ATUS, in hours per day. The model-predicted housework time is an equilibrium cross-equation check. Cost shares and expenditure ratios are fitted via the gradient-based identification of ϵ and ρ and the level identification of the remaining parameters.

The model fits the cost shares and expenditure ratios reasonably well, though not exactly, because the parameters are identified from population-weighted moments rather than from targeting individual groups. The main deviations in cost shares occur for groups 2 and 3, where the model's monotonic prediction (higher $\omega \Rightarrow$ higher s) conflicts with the non-monotonic pattern in the data. This non-monotonicity likely reflects income effects from husband's wages that the model attributes entirely to substitution.

The fit for housework time, which was not directly targeted, provides a useful model validation check. The model predicts hours ranging from 0.59 to 0.79 across groups, compared to 0.56 to 0.82 in the data. The model captures the overall level of housework time and the direction of the cross-group gradient (higher-wage women do less housework), though it overpredicts time for group 2 (wife no BA, husband BA+) by 0.10 hours per day. This group is the smallest at 7.3% of the population and exhibits atypically low housework hours given its wage level, consistent with the income-effect contamination noted above for cost shares: households in this group have high husband wages ($\lambda^g = 1.113$) relative to wife wages ($\omega^g = 0.731$), and the model attributes all cross-group variation in housework time to the wife's substitution margin, understating the role of husband's income in purchasing market substitutes. Because group 2 represents a small population share, this miss has negligible quantitative effect on the aggregate welfare and amplification results.

5 Quantitative analysis

We use the calibrated model to conduct three sets of counterfactual exercises. These exercises quantify the labor market distortion from product market power, examine its interaction with other economic forces, and draw out policy implications.

5.1 Pass-through of cost reductions

The family economics literature interprets the falling cost of home appliances, prepared foods, and other substitutes as a driver of rising female labor force participation (e.g. Greenwood, Seshadri, and Yorukoglu, 2005). That literature implicitly assumes full pass-through of costs to prices. In our framework, pass-through can exceed 100% because markups respond endogenously to cost changes, and the labor supply response to a cost reduction is correspondingly larger than standard calculations suggest. We define the *price pass-through rate* as the elasticity of the equilibrium price $r = \mu\zeta\bar{w}_m$ with respect to the cost parameter ζ:

$$\mathrm{PT} := \frac{\Delta \ln r}{\Delta \ln \zeta} = 1 + \frac{\Delta \ln \mu}{\Delta \ln \zeta} \tag{41}$$

Since $\epsilon < \sigma$, Lemma 2 implies $d\mu/d\zeta > 0$: markups fall when costs fall. Therefore $\mathrm{PT} > 1$ and cost reductions are *more* than fully passed through to prices. We reduce ζ by 10%, 25%, and 50% and compute the resulting equilibrium in both the baseline model (endogenous μ) and a frozen-markup model (μ held at its pre-shock value), defining the amplification factor as

$$\mathcal{A}_\zeta := \frac{\Delta \bar{\ell}^{\,\text{baseline}}}{\Delta \bar{\ell}^{\,\text{frozen}}} \tag{42}$$

Table 4 reports the results. The amplification factor is $\mathcal{A}_\zeta \approx 1.121$: the labor supply response to cost reductions is about 12% larger with endogenous markups than with frozen markups. A 25% reduction in ζ reduces housework time by 1.6% and raises welfare by 0.57% of consumption. The markup declines from $\mu = 1.254$ to $\mu = 1.224$[11] as demand becomes more elastic, confirming the theoretical prediction of more-than-full pass-through when $\epsilon < \sigma$. The practical implication is that standard estimates of price reductions in home inputs (see, e.g., Greenwood, Seshadri, and Yorukoglu 2005) should not be interpreted directly as a reduction in marginal cost, but rather as a combination of a falling marginal cost and an endogenous response of the markup. As a corollary, any given reduction in

[11] The equilibrium markup $\mu = 1.254$ is the value at which the model's aggregate demand satisfies the home-good firm's optimality condition; it is close to, but not identical to, the data calibration target $\mu = 1.25$ (Table 2), the sales-weighted retail markup used to identify σ. We report the equilibrium value in all counterfactuals.

marginal cost (e.g. due to a subsidy or technological progress) would lead to a 12% larger labor supply response than a competitive model would imply.

Table 4: Pass-Through of Cost Reductions

Reduction (%)	$\Delta\bar{\ell}^{\text{base}}$ (%)	$\Delta\bar{\ell}^{\text{frozen}}$ (%)	$\mathcal{A}_\zeta$	μ_{new}	CEV (%)
10	−0.6	−0.5	1.120	1.243	0.20
25	−1.6	−1.5	1.121	1.224	0.57
50	−4.3	−3.8	1.122	1.188	1.52

Note: $\Delta\bar{\ell}$ as % of baseline (40 min/day). Each row reduces ζ by the indicated fraction. "Base" denotes endogenous μ; "Frozen" holds μ at pre-shock value. $\mathcal{A}_\zeta = \Delta\bar{\ell}^{\text{base}}/\Delta\bar{\ell}^{\text{frozen}}$. CEV is the welfare gain relative to the baseline.

5.2 Amplification of wage convergence

We next quantify how endogenous markups interact with the narrowing gender wage gap. We model wage convergence as a proportional increase in female wages across all household types: for each closure fraction $f \in \{0.25, 0.50, 0.75, 1.00\}$, we set $\omega^g_{\text{new}} = \kappa\omega^g$ where

$$\kappa(f) = 1 + f\,\frac{1-\bar{\omega}}{\bar{\omega}} \tag{43}$$

and $\bar{\omega} := \sum_g n^g\omega^g$. At full closure ($f = 1$), $\kappa \approx 1.094$. We define the amplification factor as

$$\mathcal{A}_\gamma := \frac{\Delta\bar{\ell}^{\text{baseline}}}{\Delta\bar{\ell}^{\text{frozen}}} \tag{44}$$

where the baseline allows μ to adjust endogenously and the frozen-markup model holds μ at its pre-wage-change value. Since $\epsilon < \sigma$, Proposition 5 implies $\mathcal{A}_\gamma > 1$.

Table 5 reports the results. The amplification factor is $\mathcal{A}_\gamma \approx 1.012$: fully closing the aggregate gender wage gap reduces housework time by 4.32% in the baseline model, compared to 4.27% with frozen markups. The amplification is in the predicted direction but quantitatively modest, notably smaller than the cost-reduction amplification ($\mathcal{A}_\zeta \approx 1.121$). The difference arises because a reduction in ζ directly shifts the cost share s (and hence the demand elasticity faced by firms), whereas a wage change affects s only indirectly through the relative price of time.[12]

[12] Both amplification factors depend on ρ. Across $\rho \in \{0.5, 0.8, 0.90, 1.3, 1.5\}$, $\mathcal{A}_\zeta$ ranges from 1.108 to 1.266 (increasing in ρ) and $\mathcal{A}_\gamma$ from 1.039 to 1.002 (decreasing in ρ); at the estimated $\rho = 0.90$ they are

Table 5: Amplification of Wage Convergence

Closure (%)	κ	$\Delta\bar{\ell}^{\text{base}}$ (%)	$\Delta\bar{\ell}^{\text{frozen}}$ (%)	$\mathcal{A}_\gamma$	μ_{new}	$\Delta\mu$
25	1.023	−1.2	−1.1	1.012	1.252	−0.003
50	1.047	−2.3	−2.2	1.012	1.249	−0.005
75	1.070	−3.3	−3.3	1.012	1.247	−0.007
100	1.094	−4.3	−4.3	1.012	1.245	−0.010

Note: $\Delta\bar{\ell}$ as % of baseline (40 min/day). Each row raises all female wages by factor κ to close the indicated fraction of the aggregate gender wage gap $(1-\bar{\omega})$. "Base" denotes endogenous μ; "Frozen" holds μ at pre-wage-change value. $\mathcal{A}_\gamma = \Delta\bar{\ell}^{\text{base}}/\Delta\bar{\ell}^{\text{frozen}}$.

Why $A_\zeta > A_\gamma$. The contrast between $A_\zeta \approx 1.121$ and $A_\gamma \approx 1.012$ has a transparent structural explanation. To see this, consider again the representative-agent version of the model ($G = 1$, $w_m = \bar{w}_m$, $w_f = \gamma\bar{w}_m$). Let $\tau := \mu\zeta/\gamma$ denote the markup-inclusive cost of purchased home inputs relative to the wife's wage, i.e. how expensive is buying home inputs relative to doing it yourself. Note that the within-home-production price index (27) can be written as

$$Q = \gamma f(\tau), \quad \text{where } f(\tau) := \bar{w}_m \left[\phi^\epsilon + (1-\phi)^\epsilon \tau^{1-\epsilon}\right]^{\frac{1}{1-\epsilon}} \tag{45}$$

Similarly, the cost share (28) is a function of τ alone, $s = s(\tau)$. So s and Q/γ depend on (ζ, γ) only through the ratio τ. Dividing numerator and denominator of (26) by $\gamma^{1-\epsilon}$, the markup equation becomes

$$G(\tau) := \frac{\phi^\epsilon \epsilon + (1-\phi)^\epsilon \tau^{1-\epsilon}\sigma}{\phi^\epsilon + (1-\phi)^\epsilon \tau^{1-\epsilon}} = \frac{\mu}{\mu - 1} \tag{46}$$

Note that $\tau = \mu\zeta/\gamma$, so $G(\tau) = \mu/(\mu-1)$ implicitly defines μ as a function of the ratio ζ/γ alone. Since μ depends on ζ and γ only through this ratio, it follows that

$$\frac{d\ln\mu}{d\ln\zeta} = -\frac{d\ln\mu}{d\ln\gamma}, \tag{47}$$

so for log shocks of equal magnitude, the change in μ is identical across the two experiments. Since $\partial\bar{\ell}/\partial\mu$ is the same function evaluated at the same baseline, the indirect effect on $\bar{\ell}$ operating through μ is equal for both experiments. The difference in amplification therefore cannot come from the markup channel itself, but from the direct effects.

1.121 and 1.012. The wage-convergence amplification is smaller because a wage change shifts the cost share s (and hence the demand elasticity firms face) only indirectly, through the relative price of time, whereas a cost reduction shifts s directly.

To see why the direct effects differ, rewrite equilibrium housework time (32) using the τ representation. Substituting $Q = \gamma f(\tau)$ into $R = ((1-\alpha)/\alpha)^\rho Q^{1-\rho}$ and collecting powers of γ gives

$$\ell = \underbrace{\left(\frac{1-\alpha}{\alpha}\right)^\rho f(\tau)^{1-\rho}(1-s(\tau))}_{\text{depends on } \tau \text{ only}} \quad \underbrace{\gamma^{-\rho}\,\frac{1+\gamma}{\mathcal{D}(\tau,\gamma)}}_{\text{depends on } \gamma \text{ directly}} , \tag{48}$$

where $\mathcal{D} := \bar{w}_m + R[(1-s) + s/\mu]$ as in Proposition 3. Note that $\mathcal{D}$ depends on both τ (through s) and γ (through R). The cost parameter ζ enters (48) only through τ, so with μ held fixed the direct effect of ζ on ℓ operates entirely through the τ-channel. By contrast, γ enters through τ and directly through $\gamma^{-\rho}$ and $(1+\gamma)/\mathcal{D}$. Applying the chain rule with μ fixed,

$$\left.\frac{d\bar{\ell}}{d\ln\gamma}\right|_\mu = \underbrace{-\frac{\partial\bar{\ell}}{\partial\tau}\cdot\tau}_{\text{(i) }\tau\text{-channel}} + \underbrace{\left.\frac{\partial\bar{\ell}}{\partial\gamma}\right|_\tau\cdot\gamma}_{\text{(ii) direct channels}} , \tag{49}$$

where term (i) is equal and opposite to the direct effect of ζ. Term (ii) captures two additional forces that have no counterpart in the ζ experiment. First, the higher wage raises the relative price of the wife's time, which lowers her home-production time through the $\gamma^{-\rho}$ factor in (48). Second, a rise in γ raises the home production price $Q = \gamma f(\tau)$ proportionally, reallocating expenditure toward market goods when $\rho > 1$ (and toward home production at the estimated $\rho < 1$), and raises household income through the $1+\gamma$ term. In the Cobb-Douglas case $\rho = 1$, where $R = (1-\alpha)/\alpha$ is constant, (48) reduces to $\ell = K(1/\gamma + 1)$ for a constant K,[13] giving

$$\left.\frac{\partial\bar{\ell}}{\partial\gamma}\right|_\tau\cdot\gamma = -\frac{K}{\gamma} < 0, \tag{50}$$

in which the negative direct effect is the rise in the relative price of the wife's time (the $\gamma^{-\rho}$ factor), not an income effect; the income effect from the higher $1+\gamma$ works in the opposite direction, raising ℓ, and is dominated (at $\rho = 1$ it nets out, leaving only the $-K/\gamma$ substitution term). Since both terms in the γ decomposition reduce ℓ, the direct effect of γ is strictly larger in absolute value than the direct effect of ζ. Since the indirect (markup-mediated) effects are equal across experiments but the direct effect is strictly larger for the wage experiment, it follows that $A_\gamma < A_\zeta$ in the representative-agent version of the model.

In the multi-group model, $\tau^g = \mu\zeta/\omega^g$ varies across types, so the log-symmetry (47) is approximate rather than exact. Table 6 confirms that the approximation is tight. Calibrating both experiments to a shock of equal magnitude (9.4%), the "Via μ" column is nearly identical

[13] At $\rho = 1$, $R = (1-\alpha)/\alpha$ and $\mathcal{D}$ depends only on τ and μ, so $K := \frac{(1-\alpha)/\alpha\cdot(1-s(\tau))}{\mathcal{D}(\tau)} > 0$ is independent of γ.

across rows: -0.05 percentage points for wage convergence and -0.06 percentage points for the cost reduction. The markup channel adds the same absolute increment to the labor supply response regardless of the underlying shock. What differs is the "Direct" column: the frozen-markup effect of wage convergence (-4.27%) is about nine times larger than that of the cost reduction (-0.48%), driven by the income and substitution channels in term (ii) above that have no counterpart in the ζ experiment.

This decomposition clarifies how to interpret the amplification factors. The cost-reduction amplification $\mathcal{A}_\zeta \approx 1.121$ measures the markup channel against a direct effect that operates solely through the cost-share margin, where markups are active. It is the "pure" amplification of the endogenous markup mechanism. The wage-convergence amplification $\mathcal{A}_\gamma \approx 1.012$ measures the same markup channel against a much larger direct effect that includes income effects, direct substitution through $\gamma^{-\rho}$, and budget reallocation through $\mathcal{D}$. The small value of $\mathcal{A}_\gamma$ does not mean that markups respond weakly to wage convergence (the "Via μ" column shows they respond just as strongly), but rather that $\mathcal{A}_\gamma$ is small because wage convergence moves female labor supply so powerfully through non-markup channels that the (equally sized) markup contribution becomes a small fraction of the total response.

Table 6: Decomposition of Amplification Factors (9.4% Shock)

Experiment	Total $\Delta\bar{\ell}$ (%)	Direct (frozen μ)	Via μ	A
Wage convergence ($\gamma \uparrow 9.4\%$)	-4.32	-4.27	-0.05	1.012
Cost reduction ($\zeta \downarrow 9.4\%$)	-0.54	-0.48	-0.06	1.120

Note: All entries are percent of baseline $\bar{\ell}$ (40 min/day). Both experiments reduce $\bar{\ell}$ (increase female labor supply). "Direct" holds μ at its baseline value; "Via μ" is the residual from the endogenous markup response.

5.3 Welfare cost of the markup distortion

We next consider the policy-relevant question of how reducing market power would affect female labor supply and welfare. To measure welfare, we compute the consumption equivalent variation (CEV): the proportional increase in both C and X at the baseline allocation that would make the household indifferent to the counterfactual allocation. Since the utility function (1) is homogeneous of degree one in (C, X), the CEV for household type g is

$$\lambda^g = \frac{U^g_{\text{new}}}{U^g_{\text{baseline}}} - 1 \tag{51}$$

where U^g is computed from the equilibrium allocations (C^g, X^g). We report the population-weighted average $\bar{\lambda} = \sum_g n^g \lambda^g$. Welfare is computed under the model's own-group profit rebate (Section 1), so removing a markup is compensated within type and the resulting welfare change measures the deadweight loss of that markup rather than a redistribution of profits across types.[14]

Table 7 reports the results for four counterfactual scenarios. Eliminating housework substitute markups ($\mu \to 1$) reduces population-weighted housework time by 1.6% and under the own-group rebate the associated welfare gain is essentially zero. This is a second-best result driven by the small cost share of purchased inputs: with $\phi = 0.97$, purchased supplies and services are only about 9% of home-production cost, so removing the input markup, which is levied on the input price $\mu\zeta$ rather than on home output, lowers the home-good price index Q by only about 2%, and with the market markup still in place the welfare gain is negligible.[15] The time allocation effect is modest but operates in the predicted direction: removing the markup makes purchased cleaning supplies and services cheaper relative to the wife's time, shifting housework production toward purchased inputs and freeing time for market work. Eliminating the market-good markup has both a larger welfare effect (a CEV of about 0.6% of consumption, since market goods are the bulk of the budget) and, under the own-group rebate, a larger labor-supply effect (+19.7% versus +1.6% for the home markup), because the market markup taxes the entire consumption bundle. The home-substitute markup is nonetheless the novel channel of this paper: it operates through a qualitatively distinct mechanism, raising the cost of outsourcing home production and thereby directly distorting the time-allocation decision rather than the consumption margin, and it is the source of the amplification result above.

[14] Concretely, type g's income is $I^g := \tilde{\theta}^g W^g/(1 - F^g)$, where W^g is full income (valuing each spouse's time endowment, T hours per day, at their wage); $F^g := \Pi^g/I^g$ is the share of type g's income made up of the profits its own purchases generate, with Π^g defined in Section 1 (in closed form, $F^g = [\eta^{-1} + (1 - \mu^{-1})\, s^g R^g]/(1 + R^g)$, combining the market-good profit margin η^{-1} and the home-input margin $1 - \mu^{-1}$); and the scalar $\tilde{\theta}^g$ is calibrated so that the baseline reproduces the observed consumption-to-income ratio (population-weighted $\theta = 0.350$). The time-endowment scalar is common across groups and cancels in θW^g, so the counterfactuals are invariant to it. The headline labor-supply results are robust across the two profit-rebating conventions (own-group and equal per-capita); the welfare magnitudes are convention-dependent, and we flag this where it matters.

[15] Under the own-group rebate, the welfare gain from removing the home markup is close to zero for every $\epsilon \in \{1.0, 1.3, 1.55, 1.8, 2.0\}$, a second-best feature rather than an artifact of ϵ. The amplification of cost reductions does depend on ϵ: it vanishes at $\epsilon = 1$ (Cobb-Douglas, where the time-versus-input cost share is price-invariant, so the markup cannot respond to a cost change) and rises with ϵ, reaching $\mathcal{A}_\zeta \approx 1.12$ at our estimate.

Table 7: Welfare and Labor Supply Effects of Markup Distortions

Scenario	$\Delta\bar{\ell}$ (%)	CEV (%)	\$/HH/yr
No home markup ($\mu = 1$)	+1.6	≈ 0	\$0
No market markup ($\mu_M = 1$)	+19.7	0.63	\$275
Efficient (no markups)	+20.9	0.69	\$301

Note: $\Delta\bar{\ell}$ as % of baseline (40 min/day); positive means the markup increases housework time. CEV is the consumption equivalent variation of moving from the baseline to the indicated scenario. Welfare uses the own-group profit-rebate convention described at the start of Section 5. Dollar figures scale the CEV by the household's total consumption-equivalent resources ($C + X$, which include the imputed value of the wife's home-production time), not out-of-pocket expenditure.

6 Conclusion

This paper introduces monopolistic competition in home production substitutes into a model of household time allocation and estimates the resulting framework using a combination of time allocation, expenditure, and markup data. When firms that produce cleaning supplies and housekeeping services charge markups, the effective cost of outsourcing housework rises, causing households to rely more on their own time for cleaning and laundry. Because housework is disproportionately borne by women, this product market distortion reduces female labor supply. We show that the distortion interacts with other economic forces: depending on the relative magnitudes of the within-variety and across-variety elasticities of substitution, endogenous markups can amplify or dampen the labor supply effects of technological progress and gender wage convergence.

The central quantitative finding is that different brands of housework substitutes are much more easily substitutable with each other than each of them is with the household's own time. As a result, endogenous markups amplify the pass-through of cost reductions to female labor supply by approximately 12%. Standard estimates in the family economics literature, which assume perfect competition in home input markets, should be re-interpreted as measuring the combination of technological progress and endogenous markup response.

The model identifies a distinct mechanism through which product market power affects female labor supply, operating through household time allocation rather than the standard consumption margin. This mechanism has direct policy implications for regulation of housework-related industries and indirect policy implications for policies that impact the gender wage gap, since such policies can in turn influence product markups.

Several extensions are left for future work. A dynamic version could track the joint evolution of markups, technology, and female labor supply over time, enabling a formal historical decomposition (and thereby extending analyses such as Greenwood, Seshadri, and Yorukoglu 2005). The framework is also applicable to other markets, such as childcare, for which market power is pervasive and there is a link between product prices and household time allocation; the challenge is identifying the household production technology. A more complete examination of the distributional effects of market power – e.g., asking which households are most affected by home input markups – is likewise useful but would require a full-fledged heterogeneous-agent model. Finally, the framework could also be extended to incorporate labor market imperfections, providing a unified analysis of product and labor market power as determinants of gender inequality.

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A Proofs

A.1 Proofs for the household problem

Proof of Proposition 1. As stated in the main text, we focus on parameter configurations such that Equation (6) does not bind. The proof uses two-stage budgeting. At the upper level, the CES utility (1) with price indices P and Q yields expenditure shares S_C and S_X as given in (15): total spending on market goods is $S_C \cdot I$ and on home production is $S_X \cdot I$, where $I = w_m + w_f + \Pi$. At the lower levels, the demands within each nest follow from standard CES optimization and do not depend on ρ. We derive the lower-level demands below; combining them with the upper-level shares yields the proposition.

Denote by λ the Lagrange multiplier on Equation (5).

Demand for market goods. The first-order condition for c_j reads

$$\lambda p_j = \frac{\partial U}{\partial C} c_j^{-\frac{1}{\eta}} C^{\frac{1}{\eta}} \tag{52}$$

It then follows that for any $j, k \in [0,1]$,

$$\frac{p_j}{p_k} = \frac{c_j^{-\frac{1}{\eta}}}{c_k^{-\frac{1}{\eta}}},$$

or, rearranging,

$$p_j c_k^{-\frac{1}{\eta}} = p_k c_j^{-\frac{1}{\eta}}$$

Raising both sides to the power $1-\eta$, we have

$$p_j^{1-\eta} c_k^{\frac{\eta-1}{\eta}} = p_k^{1-\eta} c_j^{\frac{\eta-1}{\eta}}$$

Integrating both sides with respect to k and using the expressions for C and P in Equations (2) and (18), we get

$$p_j^{1-\eta} C^{\frac{\eta-1}{\eta}} = P^{1-\eta} c_j^{\frac{\eta-1}{\eta}}$$

Raising both sides to the power $\frac{\eta}{\eta-1}$, we get

$$p_j^{-\eta} C = P^{-\eta} c_j$$

Rearranging yields the demand function

$$c_j = \frac{p_j^{-\eta}}{P^{-\eta}} C \tag{53}$$

From Equation (52) it then also follows that

$$\lambda = \frac{1}{P} \frac{\partial U}{\partial C} \tag{54}$$

Demand for home inputs and home production time. We use a similar procedure to derive the demand for home inputs. The first-order conditions for ℓ_i and d_i, respectively, are

$$\frac{\partial U}{\partial X} \cdot x_i^{-\frac{1}{\sigma}} X^{\frac{1}{\sigma}} \cdot \phi \ell_i^{-\frac{1}{\epsilon}} x_i^{\frac{1}{\epsilon}} = w_f \lambda, \tag{55}$$

and

$$\frac{\partial U}{\partial X} \cdot x_i^{-\frac{1}{\sigma}} X^{\frac{1}{\sigma}} \cdot (1-\phi) d_i^{-\frac{1}{\epsilon}} x_i^{\frac{1}{\epsilon}} = r_i \lambda \tag{56}$$

Combining the two first-order-conditions Equations (55) and (56) gives

$$d_i = \left(\frac{1-\phi}{\phi} \cdot \frac{w_f}{r_i} \right)^{\epsilon} \ell_i \tag{57}$$

Substituting this expression into Equation (4) yields

$$d_i = (1-\phi)^{\epsilon} r_i^{-\epsilon} q_i^{\epsilon} x_i, \tag{58}$$

$$\ell_i = \phi^{\epsilon} (w_f)^{-\epsilon} q_i^{\epsilon} x_i, \tag{59}$$

where q_i is defined in Equation (16). Substituting back into the first-order conditions Equations (55) and (56), we get

$$\lambda q_i = \frac{\partial U}{\partial X} \cdot x_i^{-\frac{1}{\sigma}} X^{\frac{1}{\sigma}} \tag{60}$$

It follows that for any $i, k \in [0, 1]$,

$$\frac{q_i}{q_k} = \frac{x_i^{\frac{1}{\sigma}}}{x_k^{\frac{1}{\sigma}}},$$

or, rearranging,

$$q_i x_k^{\frac{1}{\sigma}} = q_k x_i^{\frac{1}{\sigma}}$$

Raising both sides to the power $1-\sigma$, we have

$$q_i^{1-\sigma} x_k^{\frac{\sigma-1}{\sigma}} = q_k^{1-\sigma} x_i^{\frac{\sigma-1}{\sigma}}$$

Integrating both sides with respect to k and using the expressions for X and Q in Equations (3) and (17), we get

$$q_i^{1-\sigma} X^{\frac{\sigma-1}{\sigma}} = Q^{1-\sigma} x_i^{\frac{\sigma-1}{\sigma}}$$

Raising both sides to the power $\frac{\sigma}{\sigma-1}$ gives

$$q_i^{-\sigma} X = Q^{-\sigma} x_i$$

and rearranging gives

$$x_i = \frac{q_i^{-\sigma}}{Q^{-\sigma}} X \tag{61}$$

From Equation (60), it then also follows that

$$\lambda = \frac{1}{Q} \frac{\partial U}{\partial X} \tag{62}$$

To get demand functions for d_i and ℓ_i, we substitute Equation (61) back into Equations (58) and (59) to obtain

$$d_i = (1-\phi)^{\epsilon} r_i^{-\epsilon} q_i^{\epsilon-\sigma} Q^{\sigma} X, \tag{63}$$

$$\ell_i = \phi^{\epsilon} w_f^{-\epsilon} q_i^{\epsilon-\sigma} Q^{\sigma} X \tag{64}$$

Aggregation. First, we aggregate expenditures on market goods. From Equation (53), it is immediate that

$$\int_0^1 p_j c_j \mathrm{d}j = PC \tag{65}$$

Next, we aggregate expenditures on home goods. From Equations (63) and (64), we have

$$r_i d_i + w_f \ell_i = q_i^{1-\sigma} Q^{\sigma} X$$

and so

$$\int_0^1 r_i d_i \mathrm{d}i + w_f \int_0^1 \ell_i \mathrm{d}i = QX \tag{66}$$

Adding Equations (65) and (66) and using the budget constraint Equation (5) gives

$$PC + QX = w_m + w_f + \Pi \tag{67}$$

From Equations (54) and (62), equating the two expressions for λ and using the CES utility (1) gives $QX/PC = ((1-\alpha)/\alpha)^\rho (Q/P)^{1-\rho}$. Combining with Equation (67) yields

$$C = \frac{S_C}{P}\, I, \qquad X = \frac{S_X}{Q}\, I \tag{68}$$

where S_C and S_X are the CES expenditure shares defined in (15) and $I = w_m + w_f + \Pi$. Substituting Equation (68) into Equations (53), (61), (63) and (64) yields Equations (11) to (14) in Proposition 1.

□

A.2 Proofs for the firm's problem

Proof of Proposition 2. After removing the multiplicative constants and plugging in our definition of q_i from Equation (16), the firm's problem is equivalent to solving,

$$\max_{r_i} (r_i - \zeta \bar{w}_m) r_i^{-\epsilon} \left[\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}\right]^{\frac{\epsilon-\sigma}{1-\epsilon}}$$

The firm's objective can be rewritten in log form as

$$\ln(r_i - \zeta \bar{w}_m) - \epsilon \ln(r_i) + \frac{\epsilon - \sigma}{1-\epsilon} \ln\left(\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}\right)$$

The necessary first-order condition for r_i is then

$$0 = \frac{1}{r_i - \zeta \bar{w}_m} - \epsilon \frac{1}{r_i} + (\epsilon - \sigma) \frac{(1-\phi)^\epsilon r_i^{-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}}$$

which, after multiplying through by r_i, becomes

$$0 = \frac{r_i}{r_i - \zeta \bar{w}_m} - \epsilon + (\epsilon - \sigma) \frac{(1-\phi)^\epsilon r_i^{1-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}} := F(r_i) \tag{69}$$

To show uniqueness, we follow a procedure similar to Albrecht, Phelan, and Pretnar (2025) and prove that $F'(r_i) < 0$ whenever $F(r_i) \geq 0$, where F is defined to be the right-hand side of Equation (69). Differentiating F yields

$$F'(r_i) = -\frac{\zeta \bar{w}_m}{(r_i - \zeta \bar{w}_m)^2} + (\epsilon - \sigma)(1-\epsilon) \frac{(1-\phi)^\epsilon \phi^\epsilon w_f^{1-\epsilon} r_i^{-\epsilon}}{[\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}]^2}$$

If $\epsilon > \sigma$, then $F'(r_i) < 0$ is guaranteed, so the firm's problem is concave everywhere. If $\epsilon \leq \sigma$, we note that $F(r_i) \geq 0$ implies

$$\frac{r_i}{r_i - \zeta \bar{w}_m} \geq (\sigma - \epsilon)\frac{(1-\phi)^\epsilon r_i^{1-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}} + \epsilon > (\sigma - \epsilon)\frac{(1-\phi)^\epsilon r_i^{1-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}}$$

and

$$\frac{\zeta \bar{w}_m}{r_i - \zeta \bar{w}_m} \geq (\sigma - \epsilon)\frac{(1-\phi)^\epsilon r_i^{1-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}} + \epsilon - 1 > (\epsilon - 1)\frac{\phi^\epsilon w_f^{1-\epsilon}}{\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}}$$

and so

$$r_i F'(r_i) = -\frac{\zeta \bar{w}_m r_i}{(r_i - \zeta \bar{w}_m)^2} + (\sigma - \epsilon)(\epsilon - 1)\frac{(1-\phi)^\epsilon \phi^\epsilon w_f^{1-\epsilon} r_i^{-\epsilon}}{[\phi^\epsilon w_f^{1-\epsilon} + (1-\phi)^\epsilon r_i^{1-\epsilon}]^2} < 0$$

as desired.

□

A.3 Proofs for equilibrium characterization

Proof of Proposition 3. Imposing symmetry ($c_j = C$, $x_i = X$, $d_i = d$, $\ell_i = \ell$, $P = 1$, $q_i = Q$) and using the CES upper-level allocation from Equation (15), the expenditure ratio is

$$\frac{QX}{C} = R := \left(\frac{1-\alpha}{\alpha}\right)^\rho Q^{1-\rho} \tag{70}$$

From cost minimization within X, the cost shares of time and inputs are $w_f \ell = (1-s)\, QX$ and $rd = s\, QX$, where s is defined in (28). Substituting $QX = RC$ into the time and input demands:

$$\ell = \frac{(1-s)\, R\, C}{w_f} = \frac{(1-s)\, R\, C}{\gamma \bar{w}_m} \tag{71}$$

$$d = \frac{s\, R\, C}{r} = \frac{s\, R\, C}{\mu \zeta \bar{w}_m} \tag{72}$$

Substituting into the resource constraint $\gamma \ell + C + \zeta d = 1 + \gamma$:

$$\frac{(1-s)RC}{\bar{w}_m} + C + \frac{sRC}{\mu \bar{w}_m} = 1 + \gamma$$

Solving for C:

$$C\left[1+\frac{R}{\bar{w}_m}\left((1-s)+\frac{s}{\mu}\right)\right]=1+\gamma$$

Defining $\mathcal{D}:=\bar{w}_m+R[(1-s)+s/\mu]$ and using $\bar{w}_m=(\eta-1)/\eta$, we obtain $C=\bar{w}_m(1+\gamma)/\mathcal{D}$. Substituting back into (71) and (72) yields the remaining expressions. □

A.4 Proofs of comparative statics results

Proof of Lemma 2. Equation (26) can be written as:

$$\epsilon-(\epsilon-\sigma)\frac{(1-\phi)^{\epsilon}}{\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}}=\frac{1}{1-1/\mu}$$

Fully differentiating with respect to ζ, we have

$$\frac{(\epsilon-1)(\epsilon-\sigma)\phi^{\epsilon}(1-\phi)^{\epsilon}}{[\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}]^{2}}\left(\mu^{\epsilon-2}\zeta^{\epsilon-1}\gamma^{1-\epsilon}\frac{d\mu}{d\zeta}+\mu^{\epsilon-1}\zeta^{\epsilon-2}\gamma^{1-\epsilon}\right)=-\frac{1}{(\mu-1)^{2}}\frac{d\mu}{d\zeta}$$

Multiplying both sides and rearranging, we have

$$\left[\frac{(\epsilon-1)(\sigma-\epsilon)\phi^{\epsilon}(1-\phi)^{\epsilon}}{[\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}]^{2}}\mu^{\epsilon-1}\zeta^{\epsilon-1}\gamma^{1-\epsilon}-\frac{\mu}{(\mu-1)^{2}}\right]\frac{d\mu}{d\zeta}=\frac{(\epsilon-1)(\epsilon-\sigma)\phi^{\epsilon}(1-\phi)^{\epsilon}}{[\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}]^{2}}\mu^{\epsilon}\zeta^{\epsilon-2}\gamma^{1-\epsilon} \tag{73}$$

If $\epsilon>\sigma$, the right-hand side of Equation (73) is strictly positive, and the expression in brackets on the left-hand side is strictly negative, so that $\frac{d\mu}{d\zeta}<0$. It is also straightforward to show, from Equation (73), that $\frac{\zeta}{\mu}\frac{d\mu}{d\zeta}>-1$. If $\epsilon=\sigma$, it is trivial that $\frac{d\mu}{d\zeta}=0$. If $\epsilon<\sigma$, the expression on the right-hand side of Equation (73) is strictly negative, so it remains to sign the expression in brackets on the left-hand side. From Equation (26), it follows that

$$\frac{\mu}{\mu-1}>(\sigma-\epsilon)\frac{(1-\phi)^{\epsilon}}{\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}} \tag{74}$$

and

$$\frac{1}{\mu-1}=\epsilon-1+(\sigma-\epsilon)\frac{(1-\phi)^{\epsilon}}{\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}}>(\epsilon-1)\frac{\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}}{\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}} \tag{75}$$

so that

$$\frac{(\epsilon-1)(\sigma-\epsilon)\phi^{\epsilon}(1-\phi)^{\epsilon}}{[\phi^{\epsilon}(\mu\zeta)^{\epsilon-1}\gamma^{1-\epsilon}+(1-\phi)^{\epsilon}]^{2}}\mu^{\epsilon-1}\zeta^{\epsilon-1}\gamma^{1-\epsilon}-\frac{\mu}{(\mu-1)^{2}}<0$$

so that the expression in brackets is strictly negative. It follows that $\frac{d\mu}{d\zeta}>0$ when $\epsilon<\sigma$. □

Proof of Lemma 3. The proof mirrors the proof of Lemma 2. □

B Data Appendix

B.1 Variable Dictionary

Table 8: Variable Dictionary

Variable	Description	File
NielsenIQ Consumer Panel (2019 for prices; 2007 for markups)		
`household_code`	Household identifier	Panelists
`trip_code_uc`	Shopping trip identifier; links purchases to households	Trips
`upc`, `upc_ver_uc`	Product code and version	Purchases
`total_price_paid`	Dollars paid (net of coupons)	Purchases
`quantity`	Units purchased	Purchases
`product_module_descr`	Product module description	Products
`Marital_Status`	Marital status (1 = Married)	Panelists
`Projection_Factor`	Survey weight	Panelists
Markup Estimates (Sangani, 2025)		
`Retail Markup`	Product-module-level retail markup	Scanner
`Sales Share`	Module sales share	Scanner
ATUS (2019–2024)		
`TUCASEID`	Respondent case identifier	Activity
`TRCODEP`	6-digit activity code	Activity
`TUACTDUR24`	Activity duration in minutes	Activity
`TULINENO`	Person line number within household	Respondent
`TRSPPRES`	Spouse present (1 = yes)	Respondent
`TELFS`	Labor-force status ($1, 2$ = employed; used to keep employed wives)	Respondent
`TUFNWGTP`	Final person weight	Respondent
ATUS-CPS Linked File (2019–2024)		
`PESEX`	Sex (1 = Male, 2 = Female)	CPS
`PEEDUCA`	Education (≥ 43 = BA or higher)	CPS
`PESPOUSE`	Spouse line number; used to link couples	CPS
`PRERNHLY`	Hourly earnings (2 implied decimals)	CPS
`PRERNWA`	Weekly earnings (2 implied decimals)	CPS
`PEHRUSL1`	Usual weekly hours at main job	CPS
Consumer Expenditure Survey — Interview (2024)		
`NEWID`	Consumer unit identifier	FMLI
`MARITAL1`	Marital status (1 = Married)	FMLI

Variable	Description	File
`EDUC_REF`, `EDUCA2`	Education of ref. person and spouse (≥ 15 = BA+; 14 = Associate's)	FMLI
`SEX_REF`, `SEX2`	Sex of ref. person and spouse	FMLI
`EARNCOMP`	Earner composition; mapped via `SEX_REF` to keep employed-wife CUs	FMLI
`FINLWT21`	Calibration weight	FMLI
`TOTEXPPQ`	Total expenditure, prev. quarter	FMLI
`HOUSEQPQ`	House furnishings & equipment (*not* housekeeping supplies; a market good, part of PC^g)	FMLI
`PERINSPQ`	Personal insurance & pensions (excluded from PC^g)	FMLI
`CASHCOPQ`	Cash contributions (excluded from PC^g)	FMLI
`EDUCAPQ`	Education (excluded from PC^g)	FMLI
`SHELTPQ`	Shelter: owner finance, rent, and other lodging (excluded from PC^g)	FMLI
`HLTHINPQ`	Health insurance premiums (excluded from PC^g)	FMLI
`CARTKNPQ`, `CARTKUPQ`	New and used vehicle purchases (excluded from PC^g)	FMLI
`UCC`, `COST`	Item code and expenditure (cleaning-service UCCs)	MTBI
Consumer Expenditure Survey — Diary (2024)		
`UCC 330119`	Laundry and cleaning products (supplies; part of rd^g)	EXPD
`UCC 330310`	Paper towels, napkins, and toilet tissue (part of rd^g)	EXPD
`COST`	Expenditure amount (diary week)	EXPD
`EARNCOMP`, `FINLWT21`	Earner composition and weight	FMLD

B.2 Sample Construction

Unit price (r). The unit price of housework substitutes is constructed from the 2019 NielsenIQ Consumer Panel, restricted to married households and to cleaning-subcategory modules only (see below), to match the CPS/ATUS wage window and the housework scope restriction. Each UPC is matched to the products master file via (`upc`, `upc_ver_uc`) to obtain the product module description, which is then merged with our keyword-based product classification at the module level. The average unit price r is total expenditure divided by total ounces purchased, equal to \$0.10 per ounce, computed over the ounce-measured cleaning-subcategory home-substitute purchase records from married households with positive price and quantity. Restricting to a single physical unit (the ounce, which is about 63% of cleaning-supply spending) avoids averaging dollars across incomparable package counts and sizes.

Product classification. Since computing μ_M, μ, and r requires distinguishing market goods from home production substitutes, we classify products using a rule-based procedure applied to each unique (product module, UPC) description. A set of module-specific priority rules is applied first to resolve known conflicts (for example, laundry-bleach modules are assigned to the cleaning basket even though "bleach" otherwise reads as a market good). We then assign by keywords, checking the product-module description first and falling back to the UPC description. Home substitute indicators include terms such as "frozen dinner," "ready to eat," "canned soup," "laundry detergent," and "cleaning solution," i.e., products that save cooking, cleaning, or laundry time, while market good indicators include "cigarettes," "pet food," "bottled water," "cosmetics," and "batteries," i.e., products without viable home production alternatives. Products matching neither or both are left unclassified. We then assign each product module a single label by majority vote across the classified UPCs in that module, exploiting the fact that products sharing a module (e.g., "Frozen Breakfast Food") are typically of the same type. At the product module level, 746 modules are classified as market goods and 206 as home substitutes.

We further sub-classify the 206 home-substitute modules into "food" and "cleaning" subcategories using keyword matching on the module description. Cleaning keywords include "cleaner," "detergent," "soap," "fabric," "paper towel," "sponge," "disinfectant," "laundry," "dishwash," and related terms. This yields 57 cleaning-subcategory modules (e.g., "Detergents – Heavy Duty – Liquid," "Paper Towels," "Cleaners – Bathroom," "Fabric Softeners – Liquid") and 149 food-subcategory modules. We then reclassify the cleaning-subcategory modules that are not interior-cleaning or laundry substitutes (personal care such as soap and toothpaste, automotive and outdoor products, and durable appliances such as vacuums) as market goods, so that the home-input markup and price basket matches the cleaning-supply expenditure measured in the CEX (UCCs 330119 and 330310). This leaves 46 cleaning modules. We use theseS cleaning modules for the markup μ and unit price r.

Markup moments (μ_M, μ). Product-module-level retail markup estimates are from Sangani (2025), who estimates markups from NielsenIQ scanner data covering 958 product modules in 2007. Markups with values outside $(0, 10)$ are excluded. We construct two sales-weighted average markups: $\mu_M = 1.36$ over market-good modules and $\mu = 1.25$ over cleaning-subcategory home-substitute modules. Within each category, each module's sales share is renormalized to sum to one so that the average reflects the composition of each sector rather than its aggregate size.

Wage moments ($\bar{w}_m$, ω^g, λ^g). Wage moments are constructed from the ATUS-CPS linked file, restricted to survey years 2019–2024 to match the CEX vintage. The husband is located on the CPS household roster at the wife's spouse line, and so need not himself be an ATUS respondent. Hourly wages are constructed as $\texttt{PRERNHLY}/100$ when positive, else as $(\texttt{PRERNWA}/100)\,/\,\texttt{PEHRUSL1}$ when both weekly earnings and usual hours are positive. Wages are trimmed to the interval $(\$5, \$500)$ per hour. Married couples are linked using PESPOUSE: for each woman ($\texttt{PESEX} = 2$) with a valid spouse pointer, we locate the corresponding man ($\texttt{PESEX} = 1$) in the same household using PESPOUSE as the husband's line number. Couples are retained only when the wife is the ATUS respondent and both spouses have valid wages and valid education codes. All couple and group means are weighted by the wife-respondent's ATUS final weight. The normalizing wage $\bar{w}_m$ is the weighted mean husband wage across these retained couples; by construction it equals the population-weighted average $\sum_g n^g \bar{w}_m^g$, so the normalization $\sum_g n^g \lambda^g = 1$ holds. Groups are defined by the cross of wife's and husband's educational attainment: No BA ($\texttt{PEEDUCA} < 43$) and BA+ ($\texttt{PEEDUCA} \geq 43$), yielding $G = 4$ groups. Group moments: $\omega^g = \bar{w}_f^g/\bar{w}_m$, $\lambda^g = \bar{w}_m^g/\bar{w}_m$ (husband's mean wage within group), and n^g (the weighted share of couples in group g).

Time-use moments (ℓ_g, T). Time-use moments are constructed from ATUS 2019–2024, restricted to match the CPS wage window and CEX vintage. The sample consists of employed married women ($\texttt{TRSPPRES} = 1$, $\texttt{PESEX} = 2$, $\texttt{TELFS} \in \{1, 2\}$) whose husband is also employed. We restrict to these dual-earner couples so that housework time ℓ^g, the wife's wage w_f^g, and the husband's wage w_m^g are drawn from the same population (wages are observed only for employed earners); this is the same sample used for the group moments. Housework time is the sum of daily minutes in two activity categories: interior cleaning (code 020101) and laundry (020102). Food and drink preparation (020201) and kitchen and food cleanup (020203) are excluded, consistent with the housework scope restriction. The time endowment T is average daily housework time plus average daily market work time (market work comprising codes 0501xx–0502xx); all means include zeros for respondents reporting no time in a given category on the diary day. Both means are taken over the married women linked to a husband's education (the same sample used for the group moments) and weighted by the ATUS final weight, giving $T = 5.34$ hours per day. Husband's education is linked via the CPS spouse pointer to assign each respondent to one of the four groups. Education-group moments ℓ_g are computed as the group-specific weighted mean housework hours divided by the common T.

Expenditure moments (s^g, R^g). Expenditure moments are constructed from the CEX Interview Survey (2024 PUMD release). We pool five quarterly interview files (2024Q1 through 2025Q1) and aggregate to one observation per consumer unit by averaging expenditures and weights across interviews (the CEX uses a rotating panel where the same consumer unit can appear in multiple quarterly files with different `NEWID` values, distinguished by the last digit which encodes the interview number). We restrict to married households (`MARITAL1` = 1) and assign groups using the cross of wife's and husband's education (`EDUC_REF`/`EDUCA2` ≥ 15 for BA+). The wife/husband split requires the two spouses to have different recorded sexes (`SEX2` $\neq$ `SEX_REF`); the small number of same-sex and unresolved couples are dropped, as are consumer units in which either spouse's education code is below 10 (the lowest CEX schooling categories). To match the dual-earner CPS and ATUS samples, we further restrict to consumer units in which both spouses are earners, determined from the earner-composition variable `EARNCOMP` mapped to each spouse via `SEX_REF`.

Cleaning-service expenditure. We construct cleaning-service expenditure from the MTBI (Member-level Type of Income and Expenditure) file rather than the FMLI summary variable `DOMSRVPQ`. The reason is that `DOMSRVPQ` maps to the CE hierarchical group "Personal services," which bundles three unrelated UCCs: 340906 (care for elderly/invalids), 340910 (adult day care centers), and 670320 (babysitting/childcare/daycare/preschool). It does *not* contain housekeeping services. The actual hired-cleaner UCC (340310, "Housekeeping services") is classified under "Other household expenses." We extract three cleaning-service UCCs directly from the MTBI: 340310 (housekeeping services), 340520 (household laundry/dry cleaning sent out), and 340530 (coin-operated household laundry/dry cleaning). These are aggregated to the consumer-unit–quarter level and merged with the FMLI.

Housekeeping supplies (Diary). Cleaning and laundry supplies are not collected in the CEX Interview Survey; they are recorded in the CEX Diary Survey. We therefore measure supply expenditure from the 2024 Diary, summing the "Laundry and cleaning supplies" UCCs 330119 (laundry and cleaning products) and 330310 (paper towels, napkins, and toilet tissue), which mirror the NielsenIQ cleaning basket used for μ and r. We assign Diary consumer units to the four education groups using the same rules as the Interview, average each unit's weekly supply spending across its diary weeks, weight by the Diary final weight, and annualize. Because the Interview and Diary are separate samples, the Diary supply mean is combined with the Interview cleaning-service mean at the group level (the standard CEX integration). We do *not* use the FMLI variable `HOUSEQPQ`: despite its name it is "House furnishings and equipment" (furniture, appliances, textiles), a market good that remains in PC^g, not a measure of housekeeping supplies.

Home-substitute expenditure. We measure rd^g = cleaning/laundry supplies (Diary) +

cleaning services (MTBI) (annualized), combining purchased cleaning and laundry products with hired cleaning services. Food expenditure (both food at home and food away from home) remains in market good expenditure PC^g, consistent with the housework scope restriction.

Market good expenditure is $PC^g = \texttt{TOTEXPPQ} - \text{cleaning services} - \texttt{PERINSPQ} - \texttt{CASHCOPQ} - \texttt{EDUCAPQ} - \texttt{SHELTPQ} - \texttt{HLTHINPQ} - \texttt{CARTKNPQ} - \texttt{CARTKUPQ}$ (annualized): from total Interview expenditure we subtract the hired cleaning services (which enter rd^g) together with a set of non-consumption and durable categories, so that PC^g is a current, non-durable, non-housing consumption concept. These are the non-consumption categories (personal insurance and pensions as savings, cash contributions as transfers, and education as investment) and the durable and housing categories: all shelter (`SHELTPQ`: owner-occupied housing finance, rent, and other lodging), health insurance premiums (`HLTHINPQ`), and new and used vehicle purchases (`CARTKNPQ`, `CARTKUPQ`). Stripping shelter symmetrically across owners and renters, along with vehicle purchases and the savings/transfer categories, removes the most income-elastic durable and asset-finance components of expenditure; the resulting PC^g is roughly 45–55% of total outlays. The Diary supply component of rd^g is not part of the Interview total `TOTEXPPQ`, so it is not subtracted here. The cost share is $s^g = rd^g/(rd^g + w_f^g \cdot \ell_{\text{hrs}}^g \cdot 365)$, combining CEX expenditure with CPS wages and ATUS housework time. The expenditure ratio is $R^g = (rd^g + w_f^g \cdot \ell_{\text{hrs}}^g \cdot 365)/PC^g$.